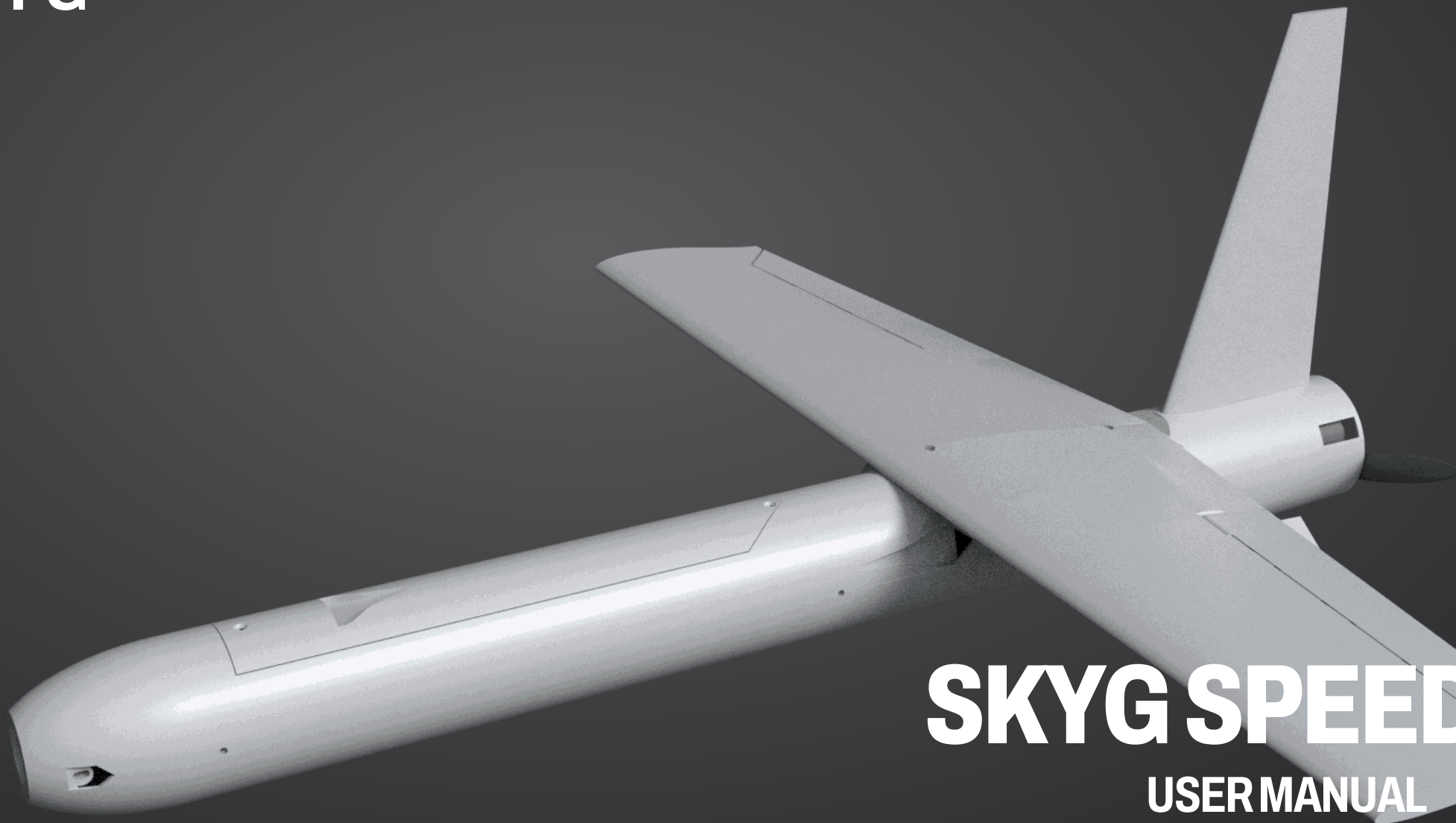


SKYG



SKYG SPEED 180

USER MANUAL

V.1 · C8-1 · 2026-10-02

PACKAGE CONTENTS

Manual

SkyG Speed 180 user manual (this document).

STEP

Every printed part and the full assembly, for editing in any CAD program.

ArduPilot

Setup document and a starting .param file for the Matek H743-MINI V3, and the rpm-governor script.

Engineering

Design record, parts list (BOM.csv), test plan and verification report.

STL

Every printed part, in millimetres, ready for the slicer (print orientation: pages in this manual).

Wiring

The wiring diagram as a separate large page.

Viewer

Interactive 3D model (open the HTML file in a browser): every part exploded and in place, with notes.

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Assembly guide

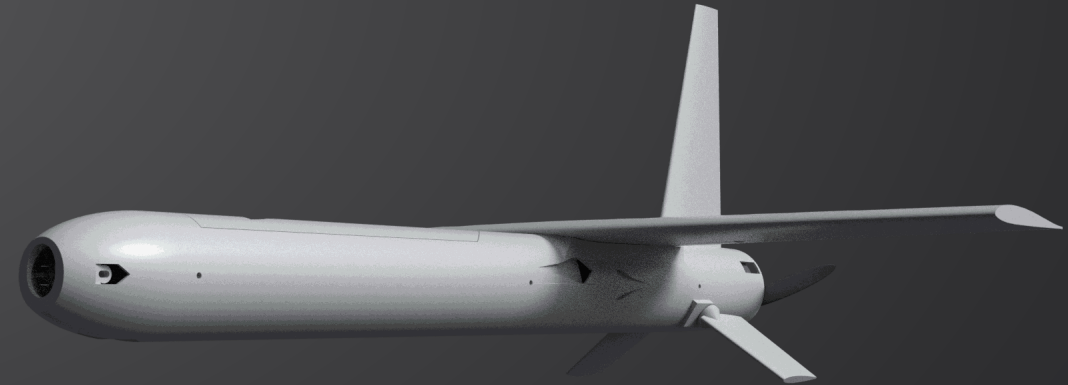
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SKYG SPEED 180

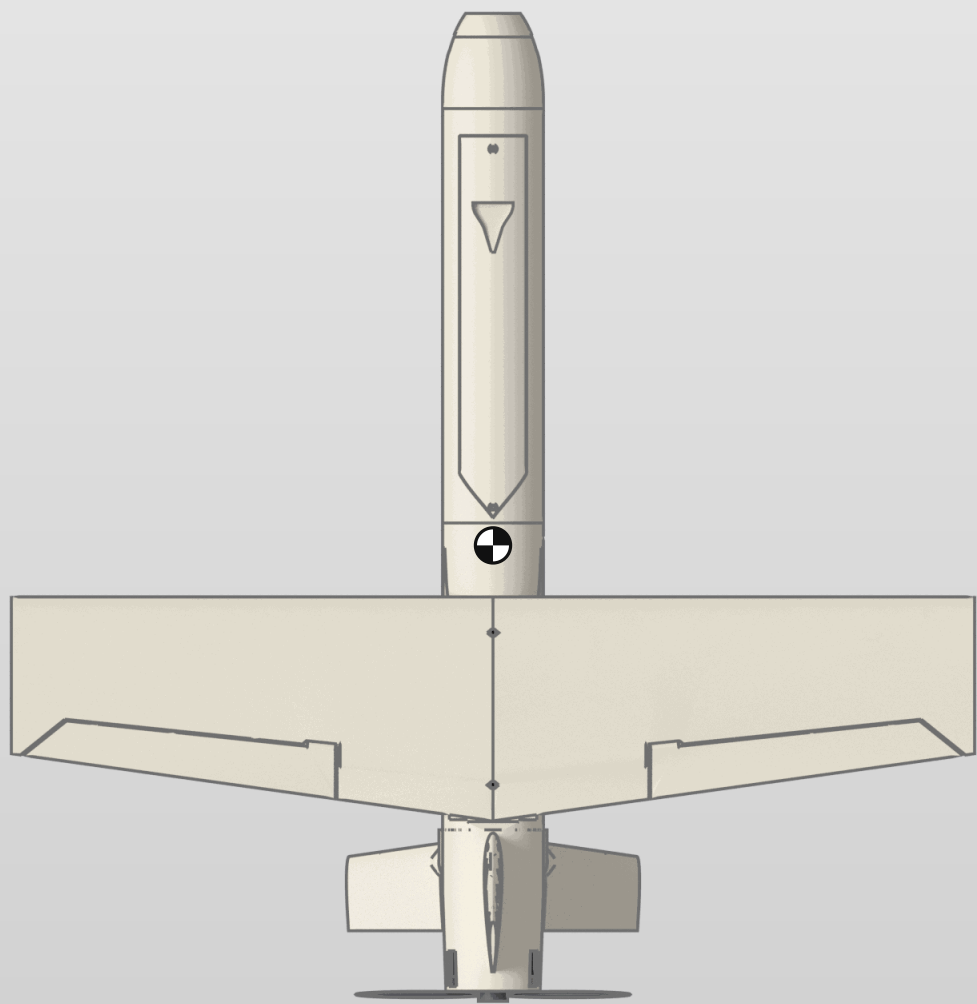
SkyG Speed 180 is a compact, hand-launched speed aircraft: a 3D-printed fuselage in three plug-together modules and a separate camera nose, a 430 mm straight wing on a carbon spar, and an all-moving canted tail driven by two micro servos. A single pusher motor sits in the tail cone behind the tail.

The airframe is printed in foaming ASA Aero on a 0.4 mm nozzle; steel and carbon parts carry the concentrated loads (spar, aileron hinge pins and torque rods, tail shafts). Six interchangeable camera noses take a micro FPV camera, a Raspberry Pi camera or a DJI O4 Air Unit, each with an open or a clear-dome face.

It is built for straight-line speed: about 162 m/s as printed and 164 m/s finished in the model with the clear-dome nose, a 12S pack and rpm-governed CNC aluminium props, flown up through never-exceed stages to 180 m/s.



GENERAL AIRCRAFT DATA



Wingspan	430 mm
Wing area	3.65 dm ²
Length	447 mm
Centre of gravity	-23 mm behind the wing leading edge at the root
All-up weight	765 g (clear-dome nose)
Stall speed	17.8 m/s
Top speed (model)	162–164 m/s
Airfoil	MH45, ailerons 3° reflex
Root / tip chord	100 / 70 mm
MAC	86 mm
Aspect ratio	5.1
Wing loading	201 g/dm ²

AERODYNAMIC DESIGN

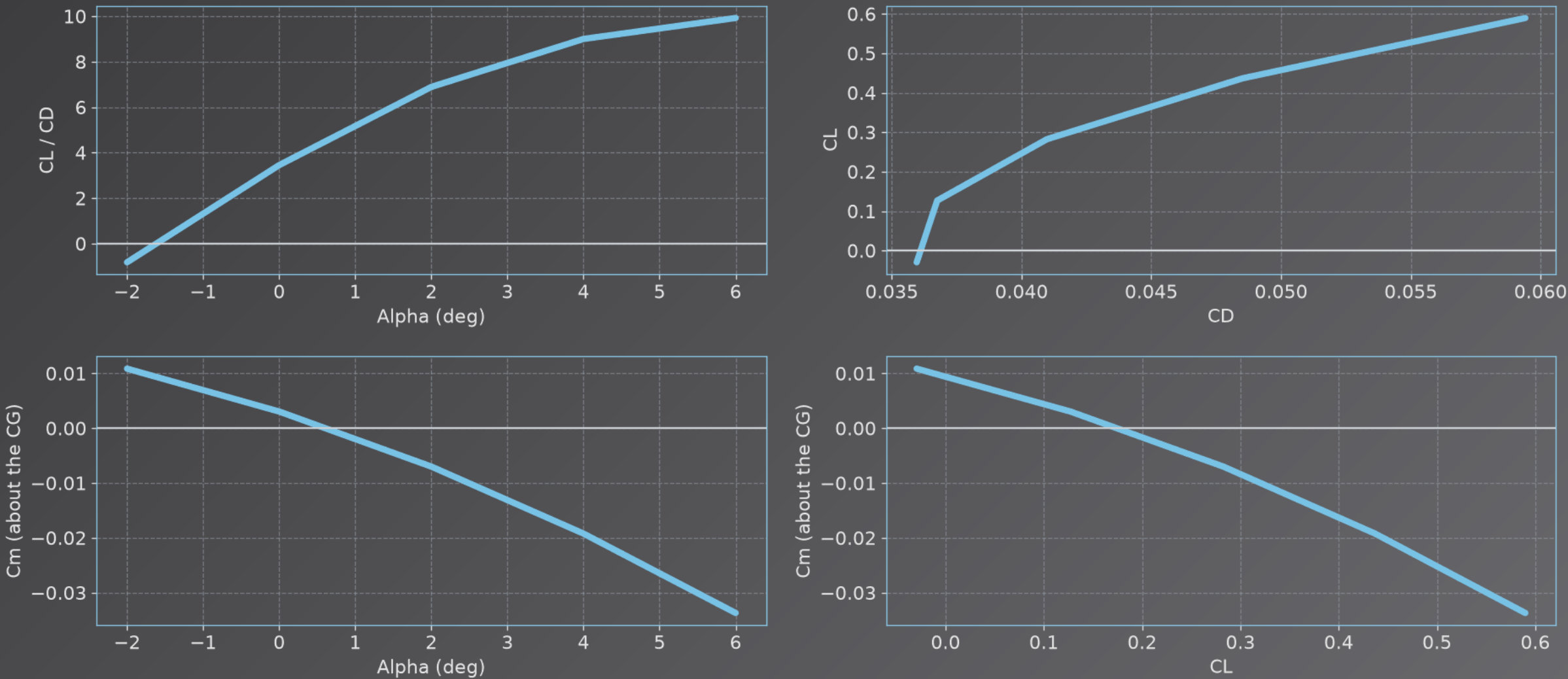
The wing uses the MH45 section, a low-moment reflexed profile, with both ailerons rigged 3° trailing edge up. The design CG is at 5 % of the mean chord, 7.0 % static margin ahead of the neutral point (x 278 mm from the nose datum): stable without a tail download penalty at speed.

Pitch and yaw come from two all-moving tail panels at 35° anhedral, mixed like a V-tail, with a fixed fin on top. The Dutch roll is lightly damped at low speed, so the flight controller's yaw damper is part of the setup.

Flutter of the wing, ailerons and tail panels was analysed (strip theory with the CAD-derived stiffness and masses): the lowest predicted instability is 223 m/s, above 1.2 × the 180 m/s top stage. The test plan opens the speed envelope in stages and measures the stiffness the margin depends on.



AERODYNAMIC CHARACTERISTICS



Whole aircraft, VSPAERO vortex lattice with the fuselage panels; best L/D 9.9 at 6.0°; C_m about the design CG.

PERFORMANCE AND SPEED STAGES

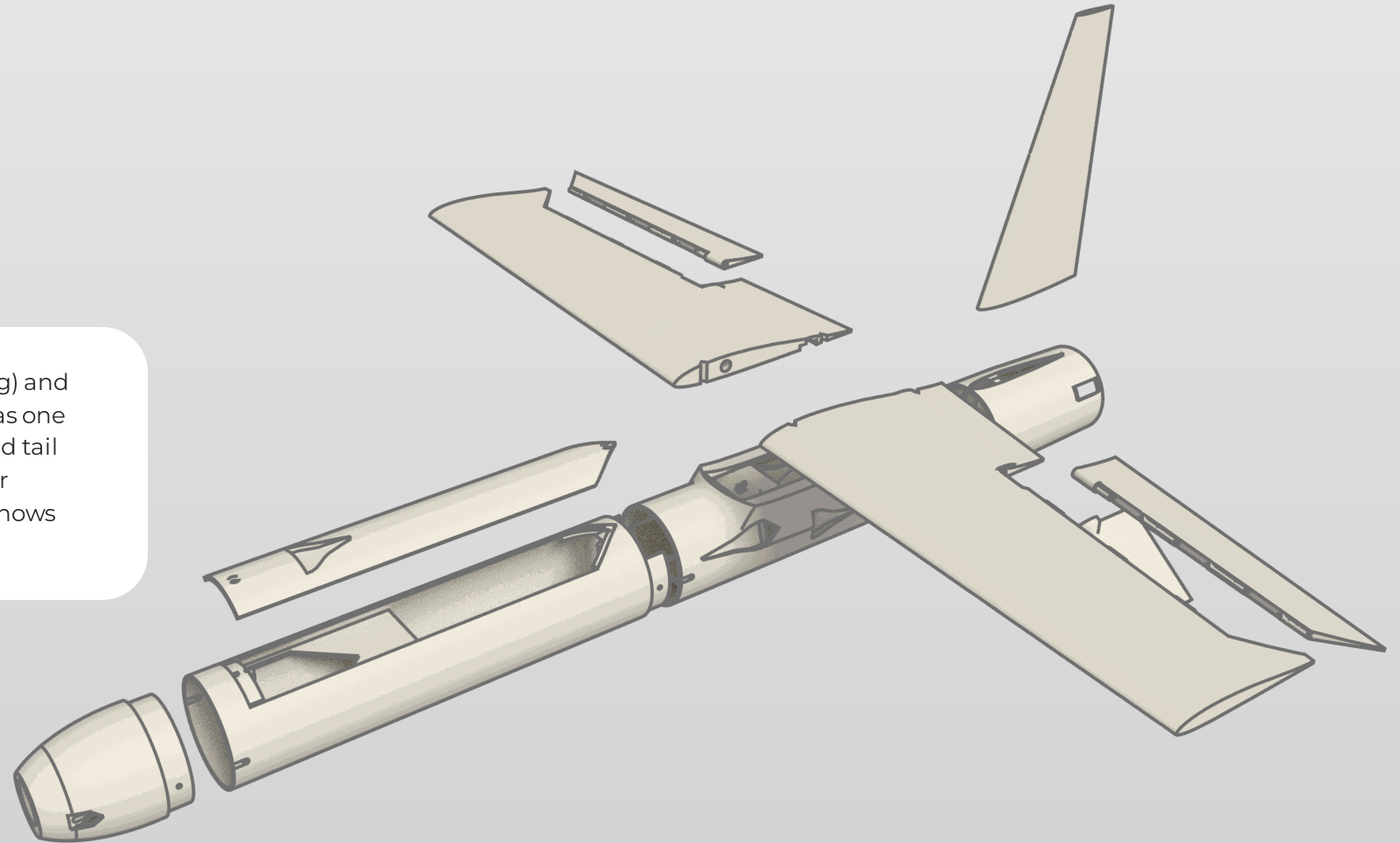
Motor	T-Motor Velox V3120 700KV
Propeller	C8-prop-P15-6x15.step and C8-prop-P18-6x18.step (CNC shop)
Battery	BetaFPV LAVA II 3S 680 95C
Static thrust	1435 g at 32 A (thrust / weight 1.88)
rpm governor / current limit	26,000 rpm max / 70 A
Top speed	162 m/s as printed, 164 m/s finished
Full-throttle time	about 0.5 min per pack pair (passes of ≤ 8 s)

Stage Vne	GOV_RPM (full stick)	Expected level speed
30 m/s	5,900 rpm	29 m/s
35 m/s	6,700 rpm	34 m/s
40 m/s	7,500 rpm	40 m/s
45 m/s	8,300 rpm	44 m/s
50 m/s	9,100 rpm	49 m/s
55 m/s	10,000 rpm	54 m/s
60 m/s	10,800 rpm	59 m/s
65 m/s	11,600 rpm	64 m/s
70 m/s	12,400 rpm	69 m/s
75 m/s	13,200 rpm	74 m/s
80 m/s	14,000 rpm	79 m/s
85 m/s	14,800 rpm	84 m/s
90 m/s	15,700 rpm	89 m/s
95 m/s	16,200 rpm	92 m/s
100 m/s	16,200 rpm	92 m/s
105 m/s	16,200 rpm	92 m/s
110 m/s	18,900 rpm	109 m/s
115 m/s	19,700 rpm	114 m/s
120 m/s	20,500 rpm	119 m/s
125 m/s	21,300 rpm	124 m/s
130 m/s	22,100 rpm	129 m/s
135 m/s	22,900 rpm	134 m/s
140 m/s	23,700 rpm	139 m/s
145 m/s	24,500 rpm	144 m/s
150 m/s	25,300 rpm	149 m/s

FILES



Every printed part is supplied as STL (for printing) and STEP (for editing), plus the complete assembly as one STEP file. Left and right wing halves, ailerons and tail panels are separate files because they are mirror images. The interactive viewer in the package shows every part, purchased item and cable in place.



COMPARTMENT LAYOUT

BATTERY BAY

12S as four 3S packs (two side-by-side pairs, one behind the other), front-loaded with the nose off; slides x 144–144 for balance

WING BAY

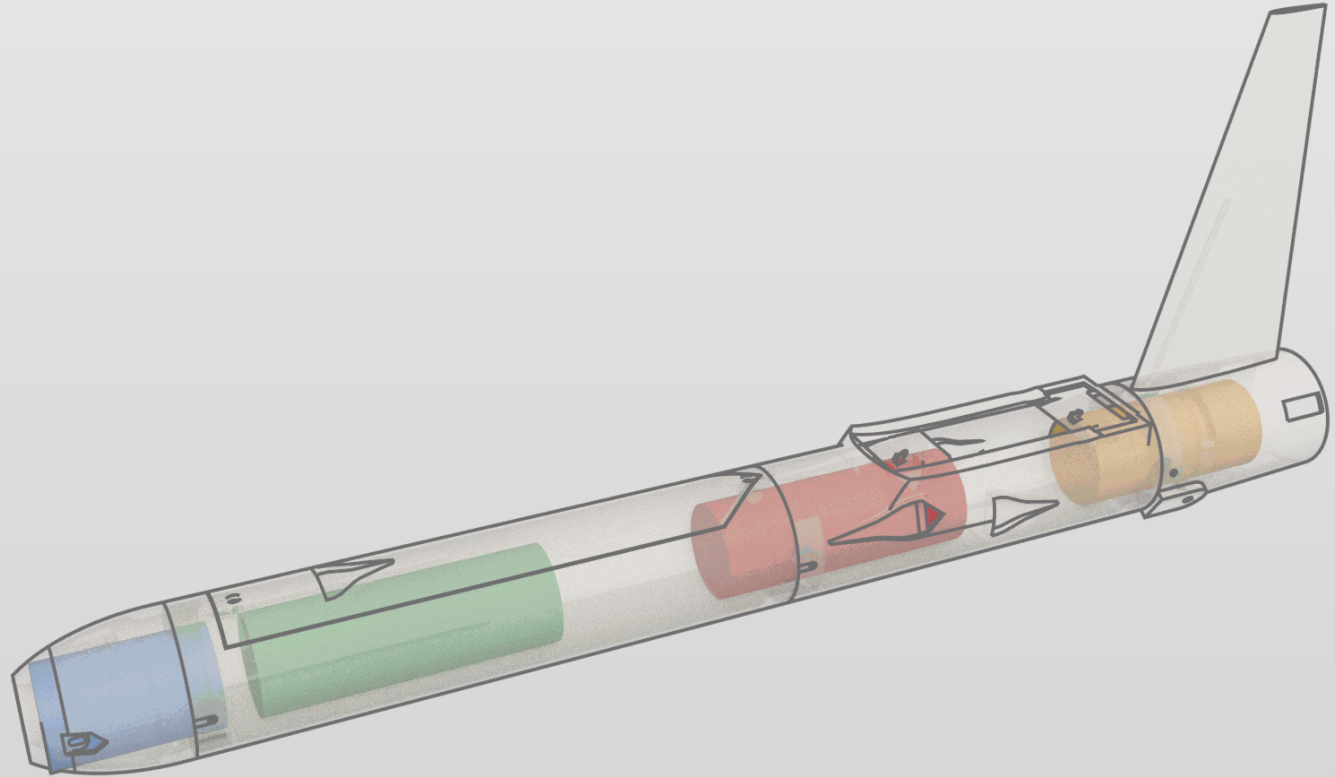
ESC, UBEC and receiver, aileron and tail servos (the flight controller is on the shelf in F1); reached with the wing off

TAIL MODULE

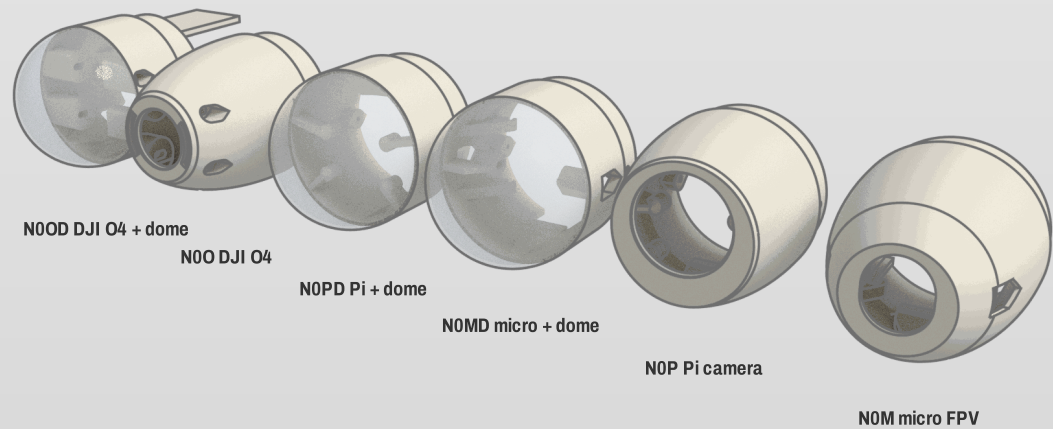
Two tail servos, pushrods, horns and the motor; plugs off at the MR30 and servo plugs

NOSE BAY

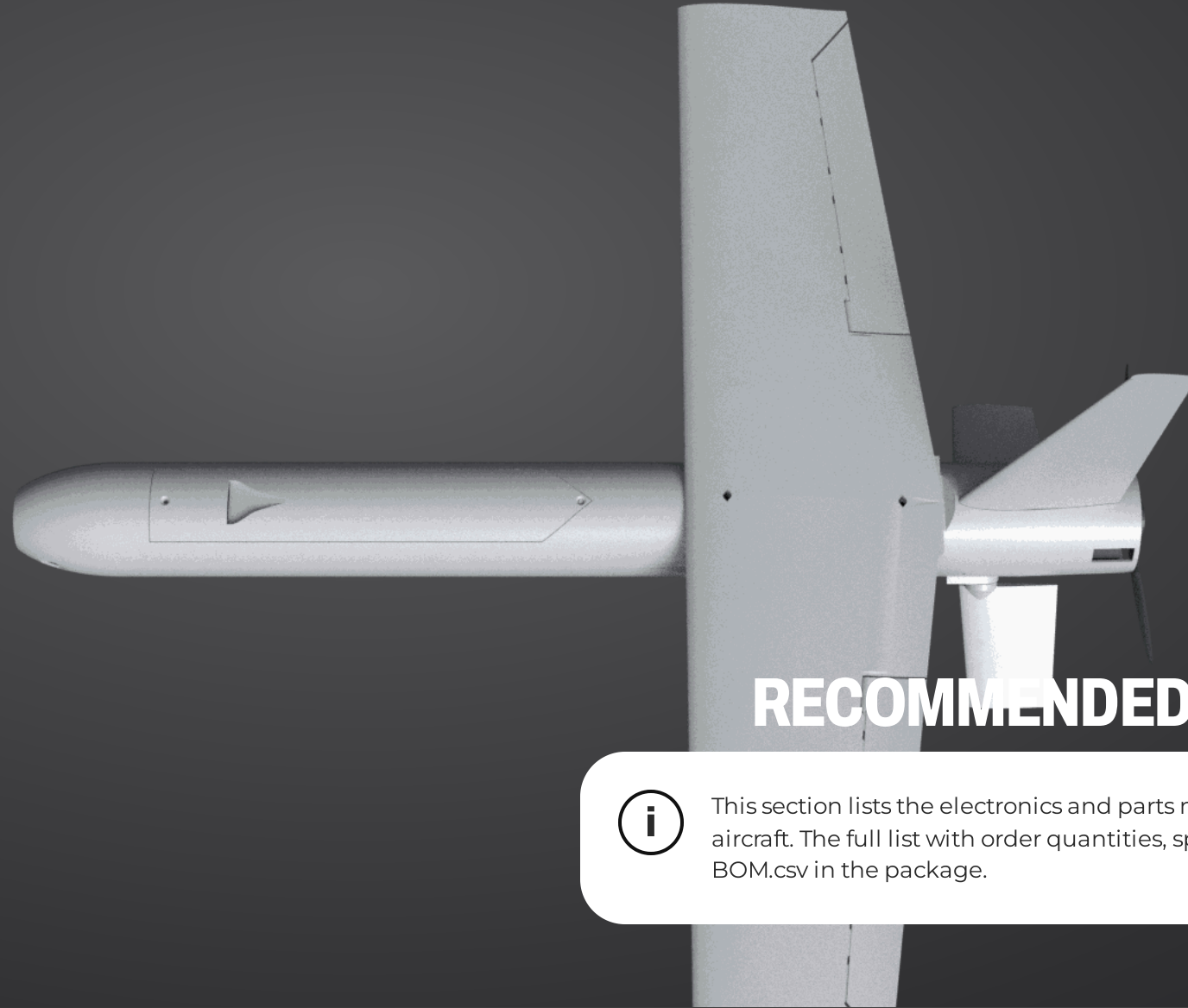
Camera and VTX (or the DJI O4 unit) in the removable nose



CAMERA NOSE VARIANTS



Nose	Camera	Battery centre for balance
N0M	19 mm micro FPV camera + 20 × 20 VTX	x 144 mm
N0P	Raspberry Pi Camera Module 3 (test option)	x nan mm (at its stop + nose ballast)
N0MD	micro camera behind a 40 mm dome, nano VTX under the hatch	x nan mm (at its stop + nose ballast)
N0PD	Pi camera behind a dome	x nan mm (at its stop + nose ballast)
N0O	DJI O4 Air Unit (Lite), ducted	x nan mm (at its stop + nose ballast)
N0OD	DJI O4 behind a dome	x nan mm (at its stop + nose ballast)



RECOMMENDED EQUIPMENT



This section lists the electronics and parts needed to complete the aircraft. The full list with order quantities, specifications and sources is BOM.csv in the package.

RECOMMENDED ELECTRONICS

COMPONENT		MODEL / SPECIFICATION
Motor		T-Motor Velox V3120 700KV
Propeller (custom CNC)		C8-prop-P15-6x15.step and C8-prop-P18-6x18.step (CNC shop)
ESC		Sequare SQESC 14130 (AM32)
Flight controller		Matek H743-MINI V3
GPS + compass		TBS M10Q GPS GLONASS (QMC5883 compass), 13.5 g
Servos		3 × Emax ES9052MD or ES9052MD HV (Promodelhobby, Turkey)
Flight battery		4 × BetaFPV LAVA II 3S 680 95C (four packs per aircraft)
Receiver		RadioMaster RP1 V2
FPV camera		Micro 19 mm FPV camera, e.g. RunCam Phoenix 2 or Caddx Ratel 2
Video transmitter + antenna		20x20 analog or digital VTX with U.FL antenna
DJI O4 Air Unit (Lite)		DJI O4 Air Unit

RECOMMENDED ACCESSORIES

ITEM	QTY
XT60 connector pairs	5
Carbon tube	1
Music wire 3/32 in (2.38 mm)	2
Tail pushrods: carbon rod + wire ends + micro ball links	2
Wheel collar for 2.5 mm shaft + horn arm	2
Nylon screw M3 x 18	1
M3 hex nut	2
M2 x 6 screws	6
M2 x 6 countersunk screws (front hatch)	2
Kapton / thin clear tape	as needed
Adhesives	as needed
Velcro strap + hook/loop	1
Heat shrink + silicone wire	as needed
Silver steel rod 2.5 mm	2
Brass tube 3 mm	2
MR30 connector pairs	1
Servo extension leads 300 mm	2

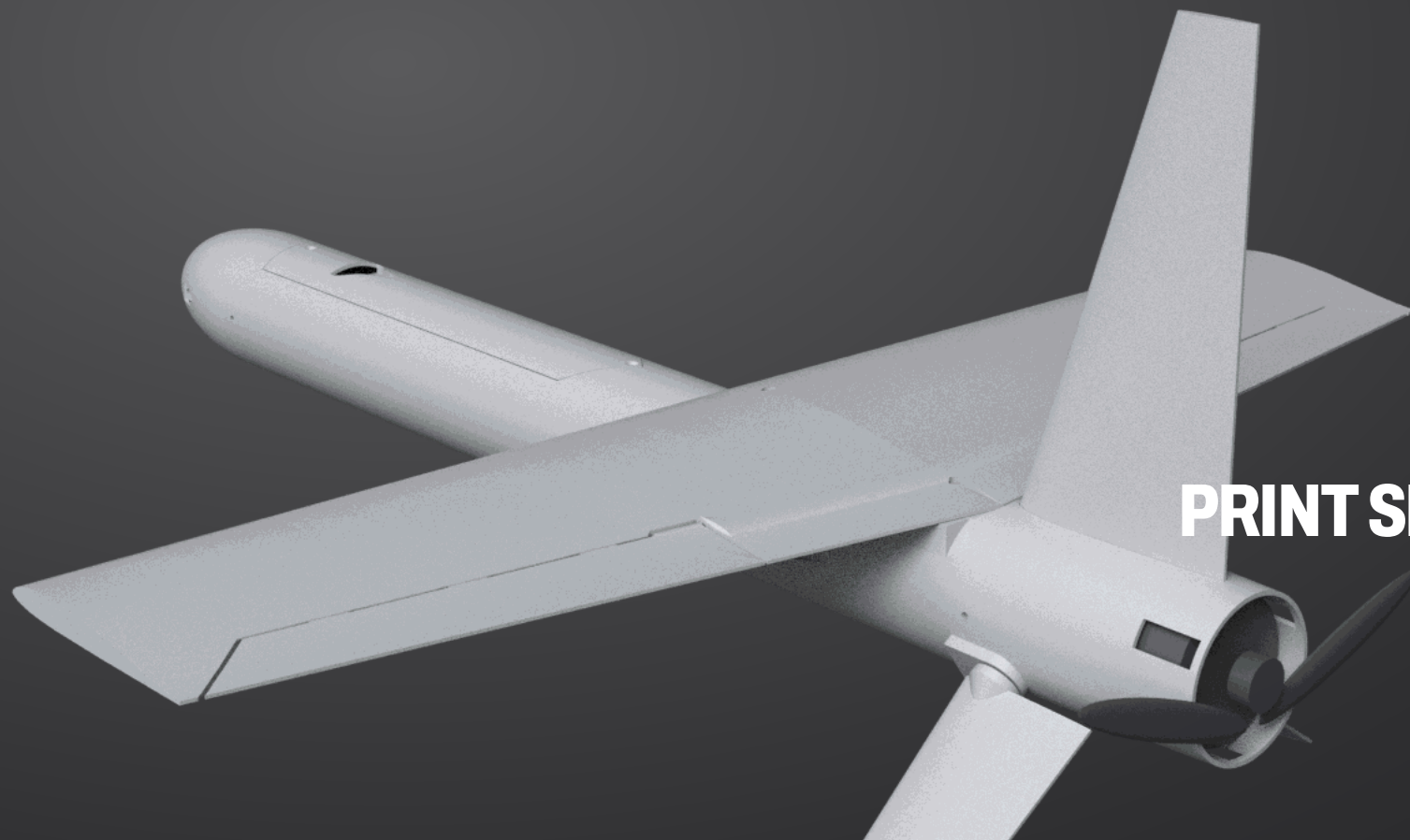
ITEM	QTY
Motor screws M3	4
Glass skin (fin and tail panels)	8 g
Nylon screw M3 x 8	1
M2 x 5 screws	4
Servo strap / zip tie + foam tape	1
Steel rod 1.5 mm	2
Vent tape	1
Music wire 0.8 mm	2
Wheel collar 3/32 in	2
15T servo arms	1
Brass sheet 1 mm	2
M1.6 nuts	2
Series harness (4 packs, 12S)	1
UBEC 4-14S dual output	1
Stainless tube 1/8 in x 0.35 mm	2
Carbon skin (wing)	30 g
Carbon rod 2 mm	2



Motor and prop. T-Motor Velox V3120 700KV (6-12S, 147 g, 37.2 mm, 38.6 mm body, 5 mm shaft, 19 x 19 M3; 79.4 A / 2453 W for 10 s; Rm 40.6 mOhm). Prop: Custom 2-blade 6 in CNC 7075-T6: P15 (6 x 15, stage 1, ~160-170 m/s) and P18 (6 x 18, stage 2, ~180 m/s); design/compact-C8/PROPS.md.

rpm governor and current limit. The flight controller runs skyg_governor.lua: it holds the prop at or below the stage rpm (never above 26,000 rpm, 92 % of the prop's rating) and the battery current at 70 A, the ESC and motor rating. In level flight the motor reaches about 23,873 rpm at full throttle; the governor acts in dives, at low speed on a full pack and per speed stage. THR_MAX 73 % is the backstop if the script stops. Prove it on the thrust stand first, and inspect the blades after every thrust-stand run and flying day.

Battery. BetaFPV LAVA II 3S 680 95C (four packs per aircraft); XT30; the two packs plug into a 2-into-1 XT30 series adapter. Full throttle in bursts of ≤ 10 s, then ≥ 30 s at low throttle: the ESC lives on its thermal mass. Static thrust in the model: 1435 g at 32 A; confirm on a thrust stand before the first flight.



PRINT SETTINGS

PRINT SETTINGS



The airframe is designed for **Bambu ASA Aero** (foaming, lightweight, heat deflection 85 °C) on a **0.4 mm nozzle** and **0.20 mm layers**; every part fits a Bambu H2S or H2D build volume. The motor plate and spinner are PETG-CF. ASA needs the closed chamber (45–60 °C), a dried spool (80 °C, 8 h) and glue on the plate; print long thin parts (wing halves, ailerons) with a brim against warping.

GENERAL GUIDELINES

- Fuselage, nose and hatch shells: 2 walls, no infill (the internal ribs and bosses are modelled).
- Wing halves and ailerons: 1 wall, 5 % gyroid.
- Tail panels: 2 walls, 10 % gyroid. Fin: 2 walls, 15 %.
- Motor plate and spinner: PETG-CF, 4 walls, 40 %.
- Tall standing parts: 5 mm brim, slow outer wall, enclosure door closed.

SUPPORTS: OFF

Every part is designed to print without supports in the orientation shown (DFAM rules, print audit: all parts pass). Keep supports switched off. The only bridge longer than 12 mm is under the rear wing-screw nut in F2, hidden inside the wing bay: if it sags on your printer, paint a small support under it there. Check the top of the aileron's torque-rod bore on the first aileron.

PRINT SETTINGS

FIRST PART AND WEIGHT CHECK

Print F1 first and weigh it. If it is more than 15 % over the sliced mass (about 18 g), lower the ASA Aero temperature or flow before printing the rest; foaming filaments vary with temperature, humidity and the enclosure.

COUPON TESTS

Before the full build, print the last 25 mm of F1 with the first 25 mm of F2, and the F2/F3 and nose joints the same way: every joint must slide on and off by hand and the M2 screws must bite. Print a 40 mm wing-root section and check the spar fit.

SLICER

The masses in this manual come from Bambu Studio with the H2S 0.4 mm profile and the walls/infill listed on the orientation pages. Import the STL files in millimetres; keep each part in the orientation shown.

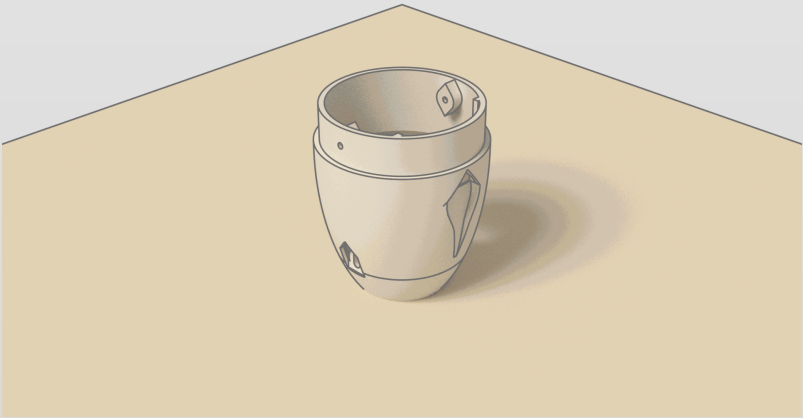
PARTS LIST

PART	MATERIAL	MASS
N0M NOSE (MICRO FPV)	ASA Aero	4.9 g
N0P NOSE (PI CAMERA)	ASA Aero	4.6 g
N0MD NOSE (MICRO + DOME)	ASA Aero	3.1 g
N0PD NOSE (PI + DOME)	ASA Aero	3.4 g
N0O NOSE (DJI O4)	ASA Aero	6.1 g
N0OD NOSE (DJI O4 + DOME)	ASA Aero	4.5 g
F1 FUSELAGE FRONT	ASA Aero	18.4 g
F2 FUSELAGE MAIN	ASA Aero	14.4 g
F3 TAIL MODULE	ASA Aero	6.5 g
HF FRONT HATCH	ASA Aero	3.2 g
FN FIN	ASA Aero	6.2 g
VTX SHELF (N0M)	ASA Aero	0.4 g
DOMR DOME RING	ASA Aero	0.4 g
HG GPS CARRIER	ASA Aero	0.7 g

PART	MATERIAL	MASS
WR WING HALF RIGHT	ASA Aero	11.0 g
WL WING HALF LEFT	ASA Aero	11.6 g
AR AILERON RIGHT	ASA Aero	1.5 g
AL AILERON LEFT	ASA Aero	1.3 g
SR TAIL PANEL RIGHT	ASA Aero	2.2 g
SL TAIL PANEL LEFT	ASA Aero	2.1 g
M1 MOTOR PLATE	PETG-CF	3.1 g

One aircraft with the N0M nose: 88 g printed. Only one camera nose is needed; the others are options.

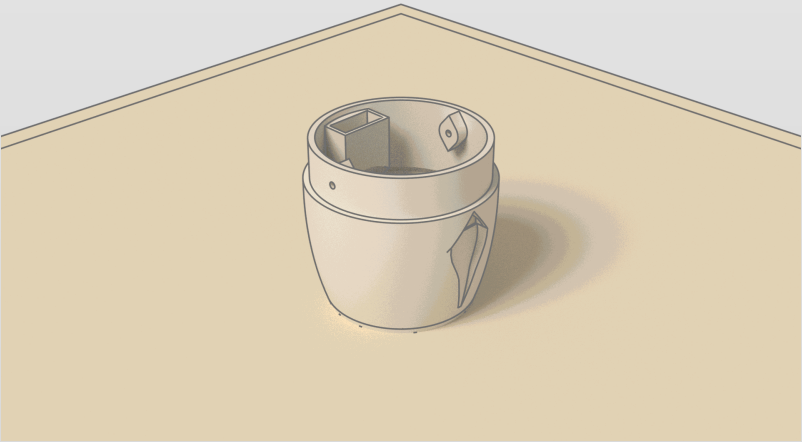
PARTS ORIENTATION



N0M NOSE (MICRO FPV)

FACE DOWN, SPIGOT UP

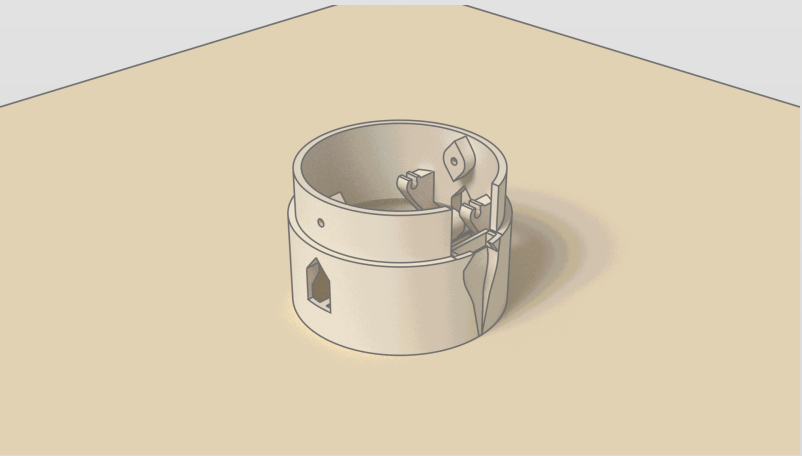
2 WALLS · 0 % INFILL · ASA AERO



N0P NOSE (PI CAMERA)

FACE DOWN, SPIGOT UP

2 WALLS · 0 % INFILL · ASA AERO

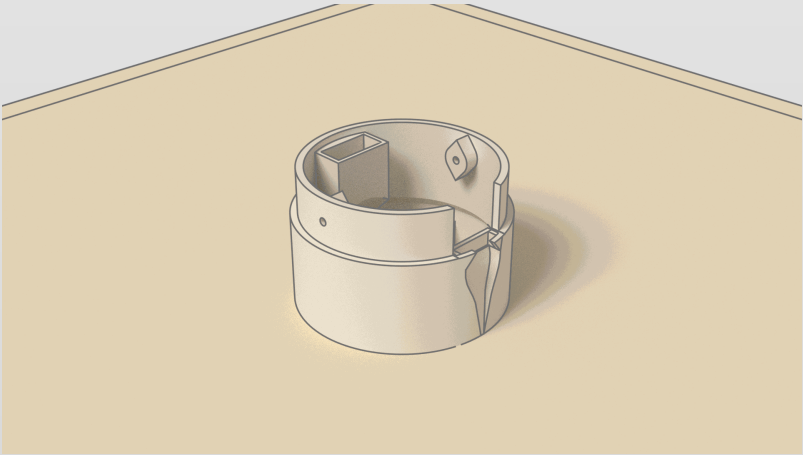


N0MD NOSE (MICRO + DOME)

FRONT FACE DOWN

2 WALLS · 0 % INFILL · ASA AERO

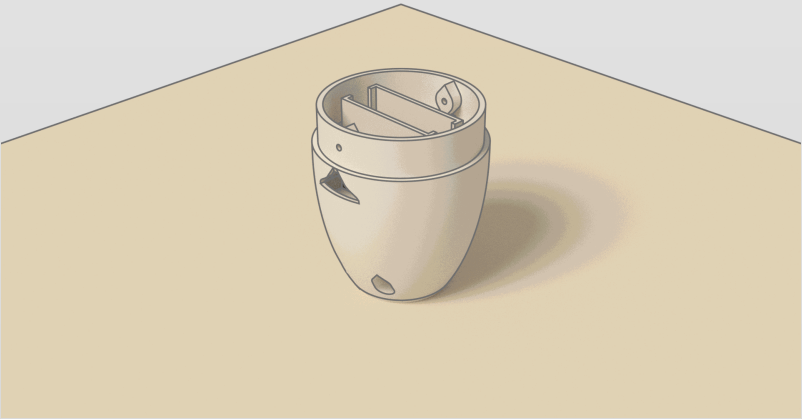
PARTS ORIENTATION



N0PD NOSE (PI + DOME)

FRONT FACE DOWN

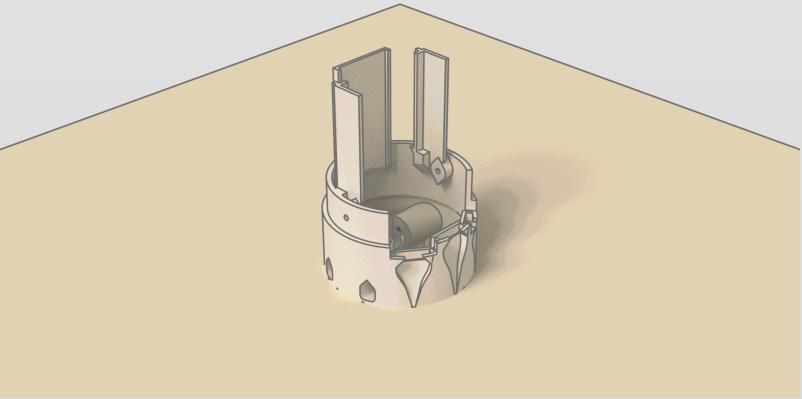
2 WALLS · 0 % INFILL · ASA AERO



N00 NOSE (DJI O4)

FACE DOWN

2 WALLS · 0 % INFILL · ASA AERO

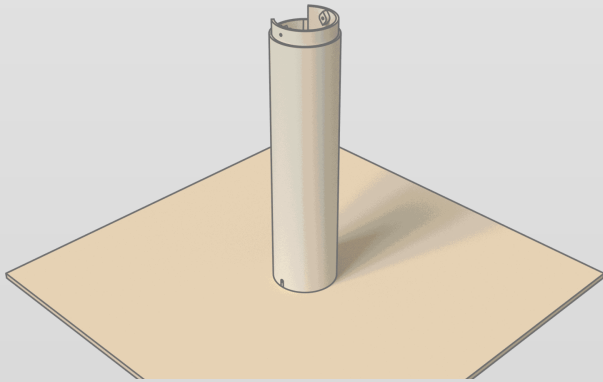


N00D NOSE (DJI O4 + DOME)

FRONT FACE DOWN

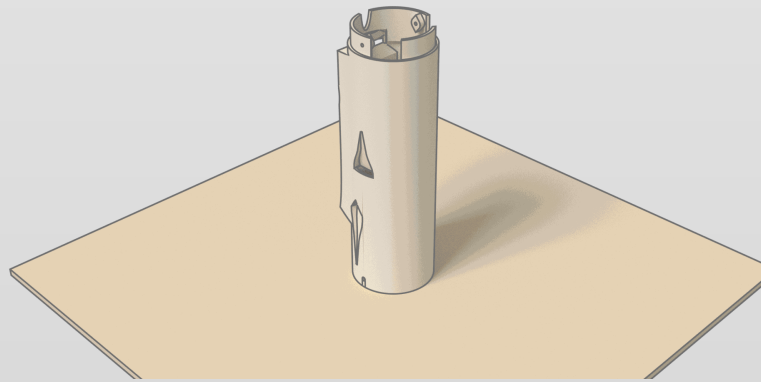
2 WALLS · 0 % INFILL · ASA AERO

PARTS ORIENTATION



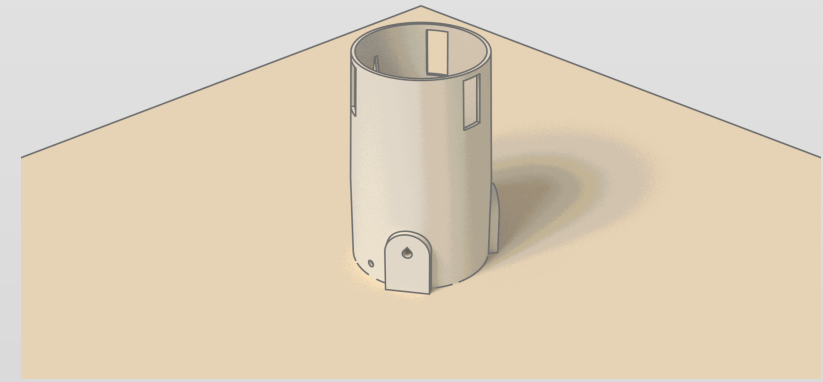
F1 FUSELAGE FRONT

FRONT END DOWN · SPIGOT UP · 5 mm BRIM
2 WALLS · 0 % INFILL · ASA AERO



F2 FUSELAGE MAIN

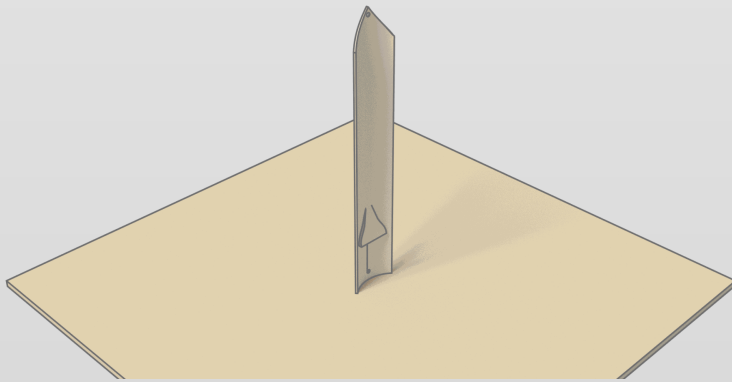
FRONT END DOWN · SPIGOT UP · 5 mm BRIM
2 WALLS · 0 % INFILL · ASA AERO



F3 TAIL MODULE

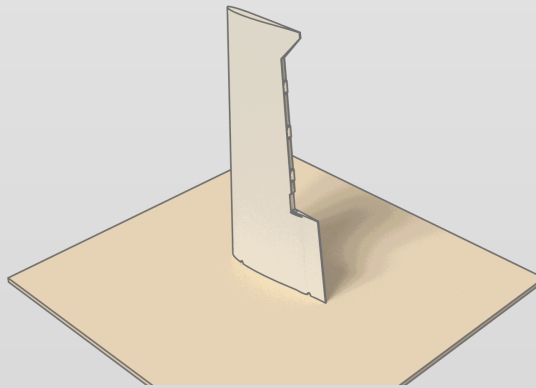
FRONT END (SOCKET) DOWN · 5 mm BRIM
2 WALLS · 0 % INFILL · ASA AERO

PARTS ORIENTATION



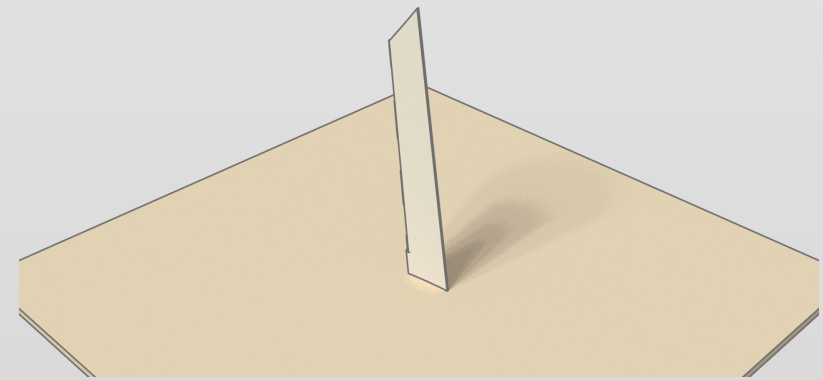
HF FRONT HATCH

STANDING, FRONT END DOWN
2 WALLS · 0 % INFILL · ASA AERO



WR WING HALF R/L

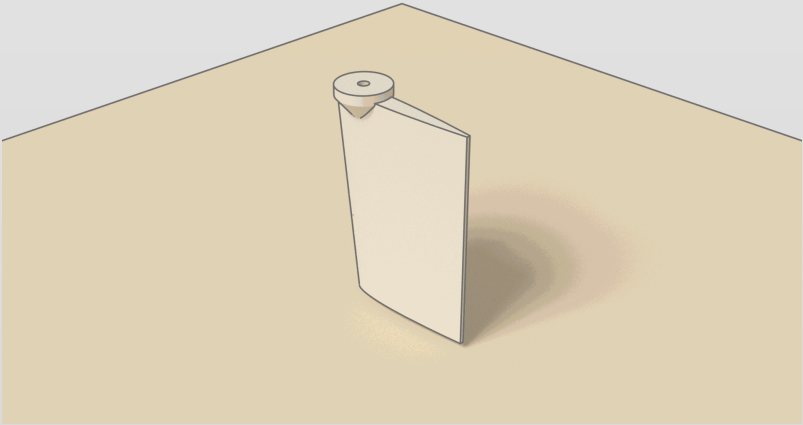
ON THE ROOT RIB
1 WALLS · 5 % INFILL · ASA AERO



AR AILERON R/L

ON THE INBOARD END
1 WALLS · 5 % INFILL · ASA AERO

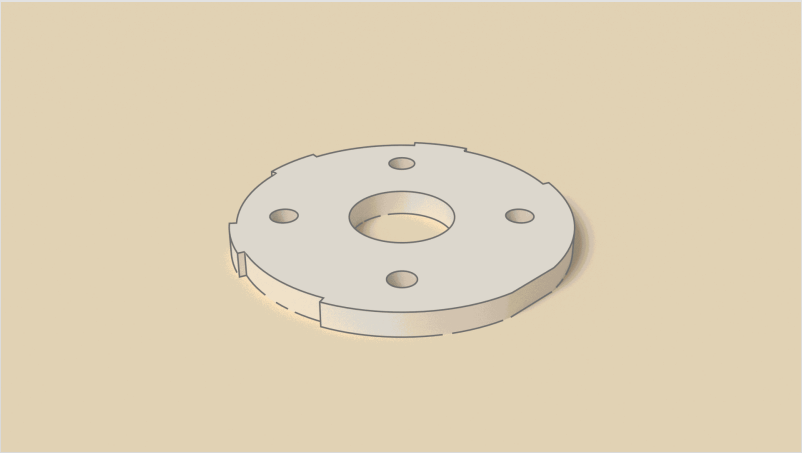
PARTS ORIENTATION



SR TAIL PANEL R / L

TIP DOWN, HUB UP

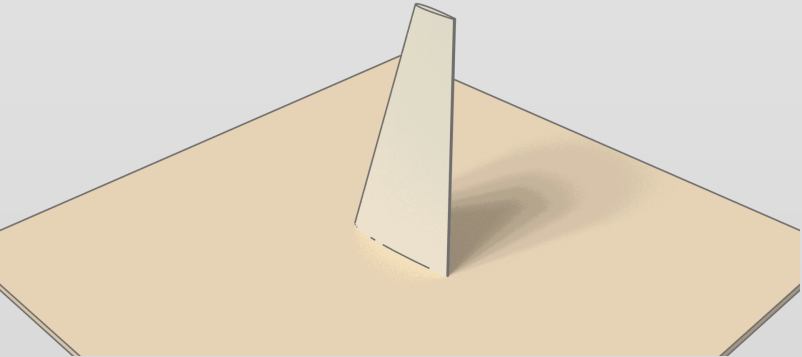
2 WALLS · 10 % INFILL · ASA AERO



M1 MOTOR PLATE

FLAT

4 WALLS · 40 % INFILL · PETG-CF

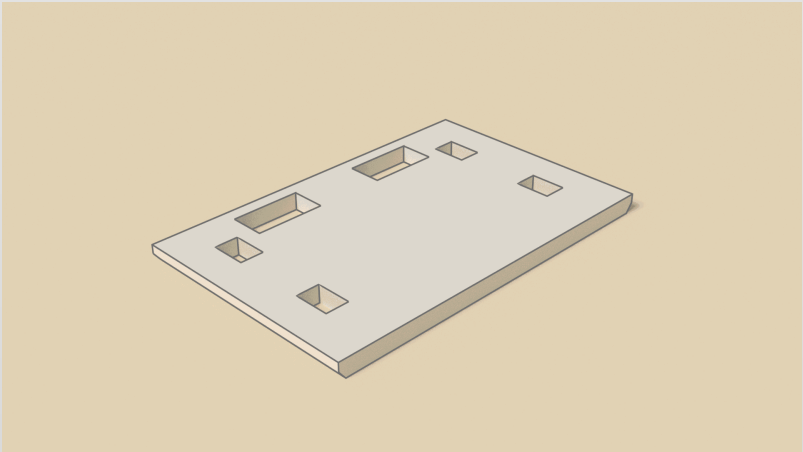


FN FIN

ON THE ROOT TAB

2 WALLS · 15 % INFILL · ASA AERO

PARTS ORIENTATION



VTX SHELF (NOM)

FLAT

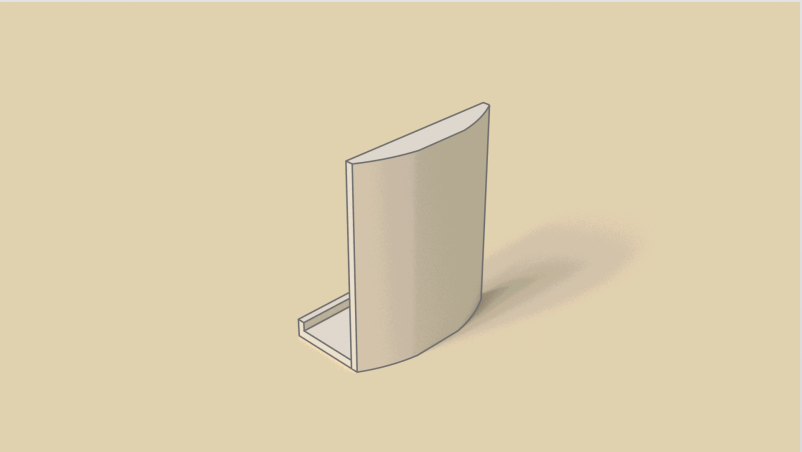
2 WALLS · 15 % INFILL · ASA AERO



DOMERING

LIP DOWN

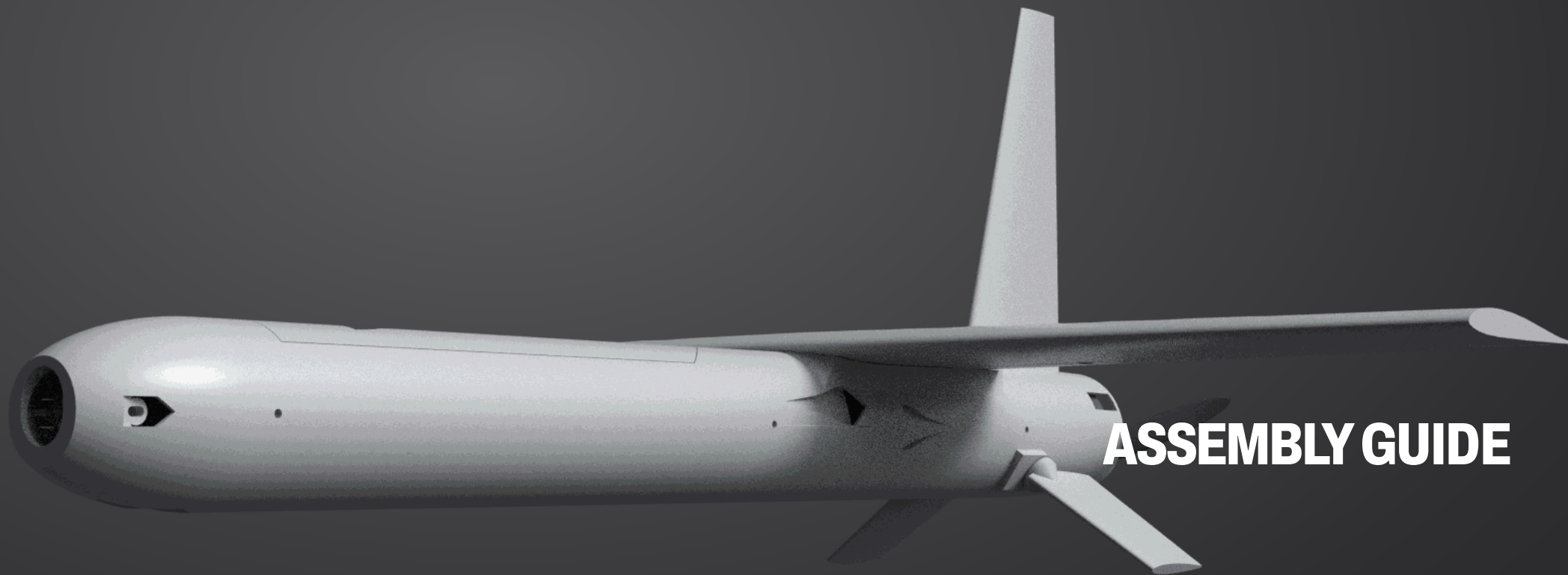
2 WALLS · 15 % INFILL · ASA AERO



HG GPS CARRIER

ON ITS REAR WALL

2 WALLS · 15 % INFILL · ASA AERO

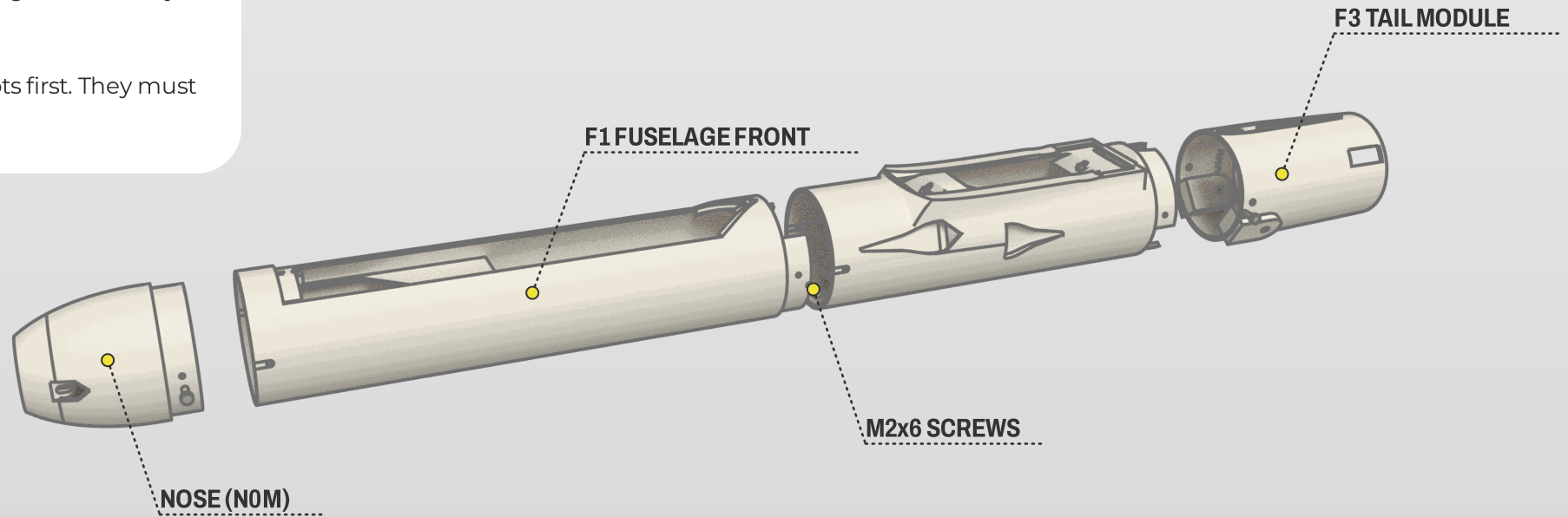


FUSELAGE ASSEMBLY



The fuselage is four modules on spigot joints: the camera nose, F1, F2 and the F3 tail module. Each joint is held by two M2 × 6 screws from the sides into bosses inside the spigot; nothing is glued, so every module comes off for service.

Clean any stringing from the spigots first. They must slide in by hand.

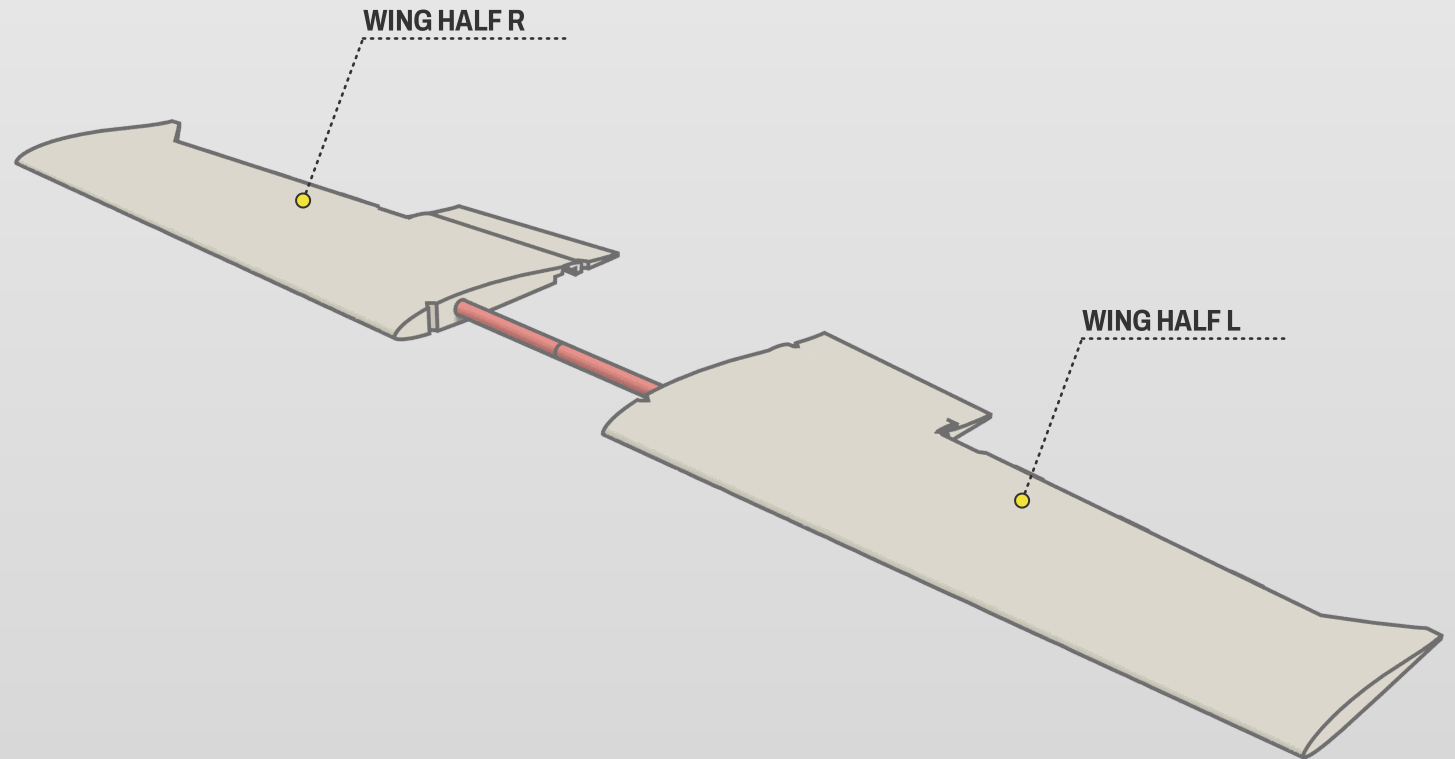


M2 × 6 SCREWS

WING ASSEMBLY



Cut one 400 mm length of 5 × 3 mm carbon tube and slide it through both wing halves (200 mm into each). Epoxy the tube in the bores and the two root ribs together with the whole wing flat on a table (0° dihedral).

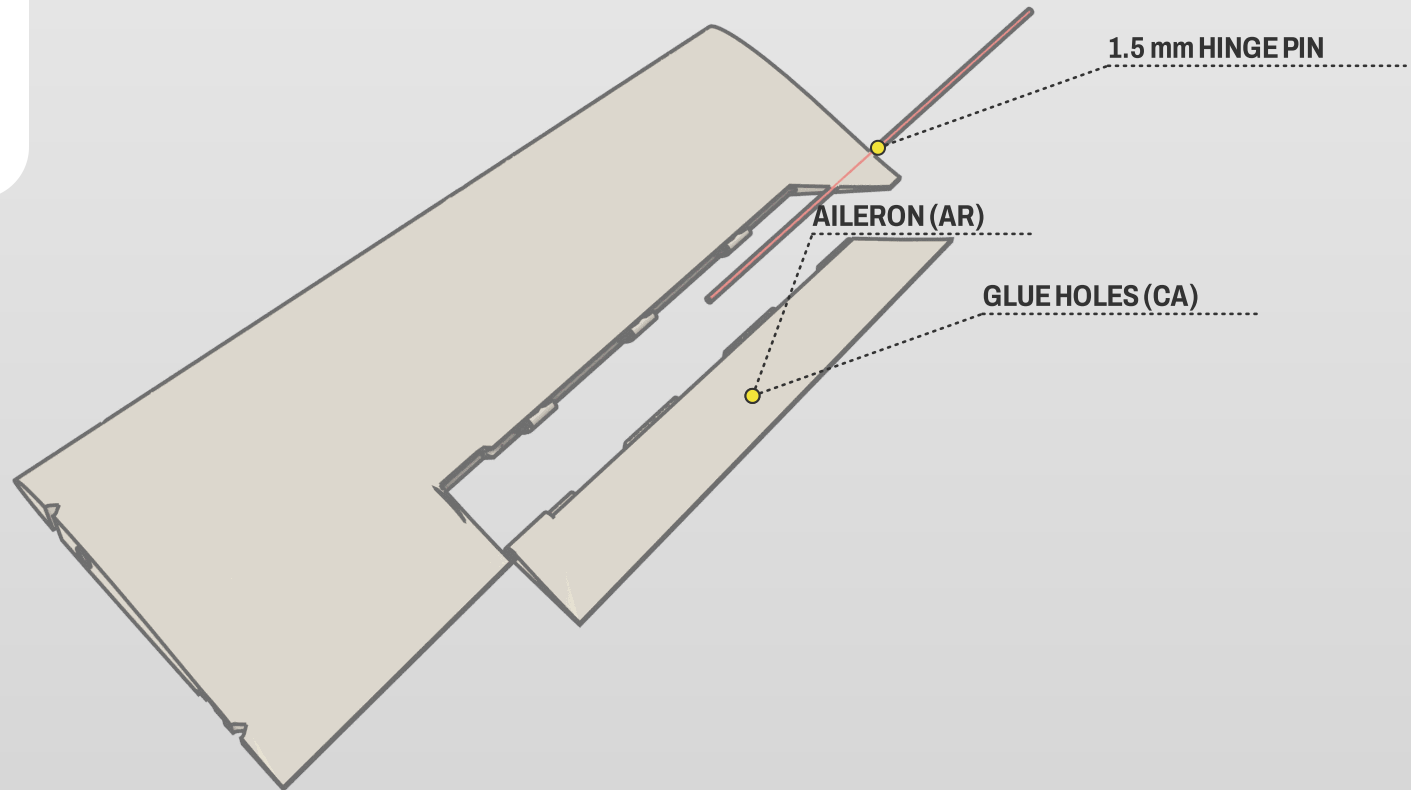


EPOXY

WING ASSEMBLY



Set each aileron in place: its rounded nose sits in the cove along the wing's hinge face and the wing's three knuckles enter the pockets in the nose. Push a 132 mm length of 1.5 mm steel rod in from the wingtip through the tip, the aileron and the knuckles. Wick thin CA into the aileron's four glue holes only, then plug the tip hole. The aileron must swing freely.

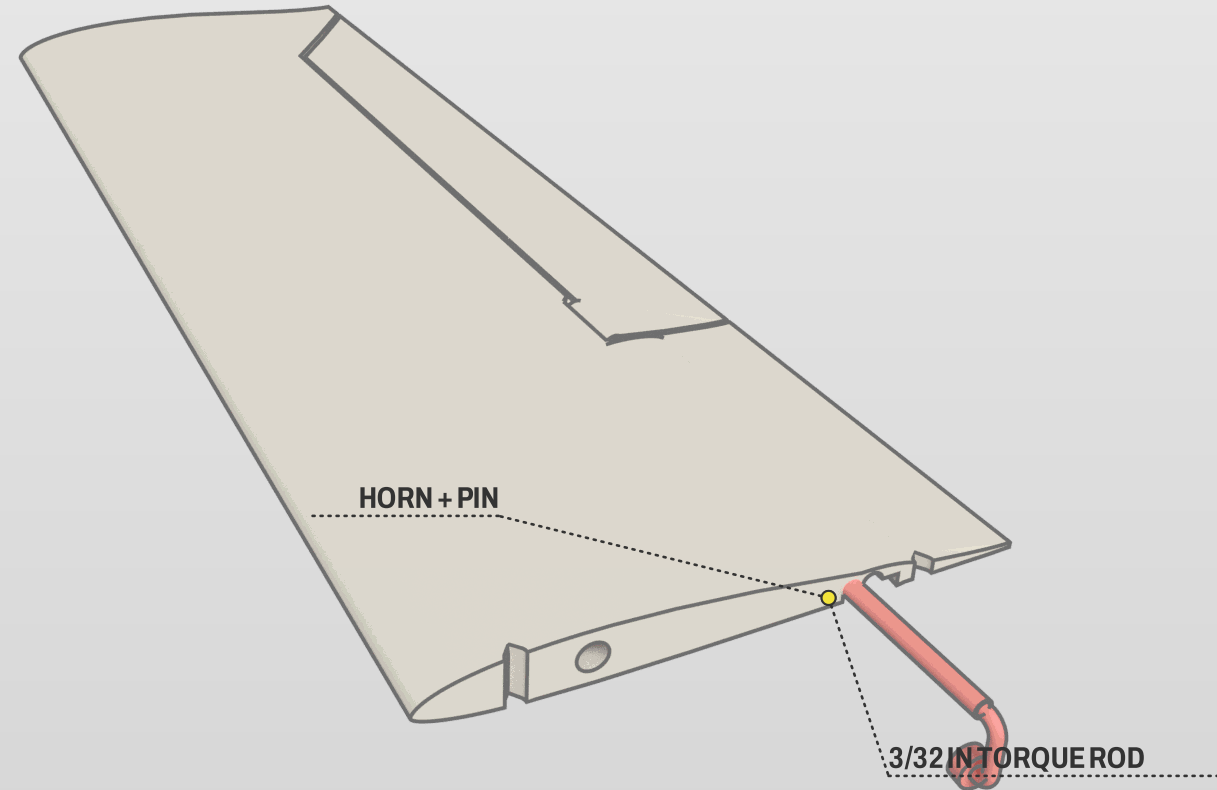


THIN CA

WING ASSEMBLY



Epoxy the stainless sleeve over each rod first (y 6–68 mm). Bend a 12 mm horn down at one end of each 3/32 in (2.38 mm) rod with a 5.5 mm pin outboard at its tip, and file a 0.3 mm flat on the last 9 mm of the straight end, opposite the horn. Push it in from the root face until the flat enters the D-bore of the aileron's root knuckle; epoxy the end and hold the aileron 3° trailing edge up while it cures.

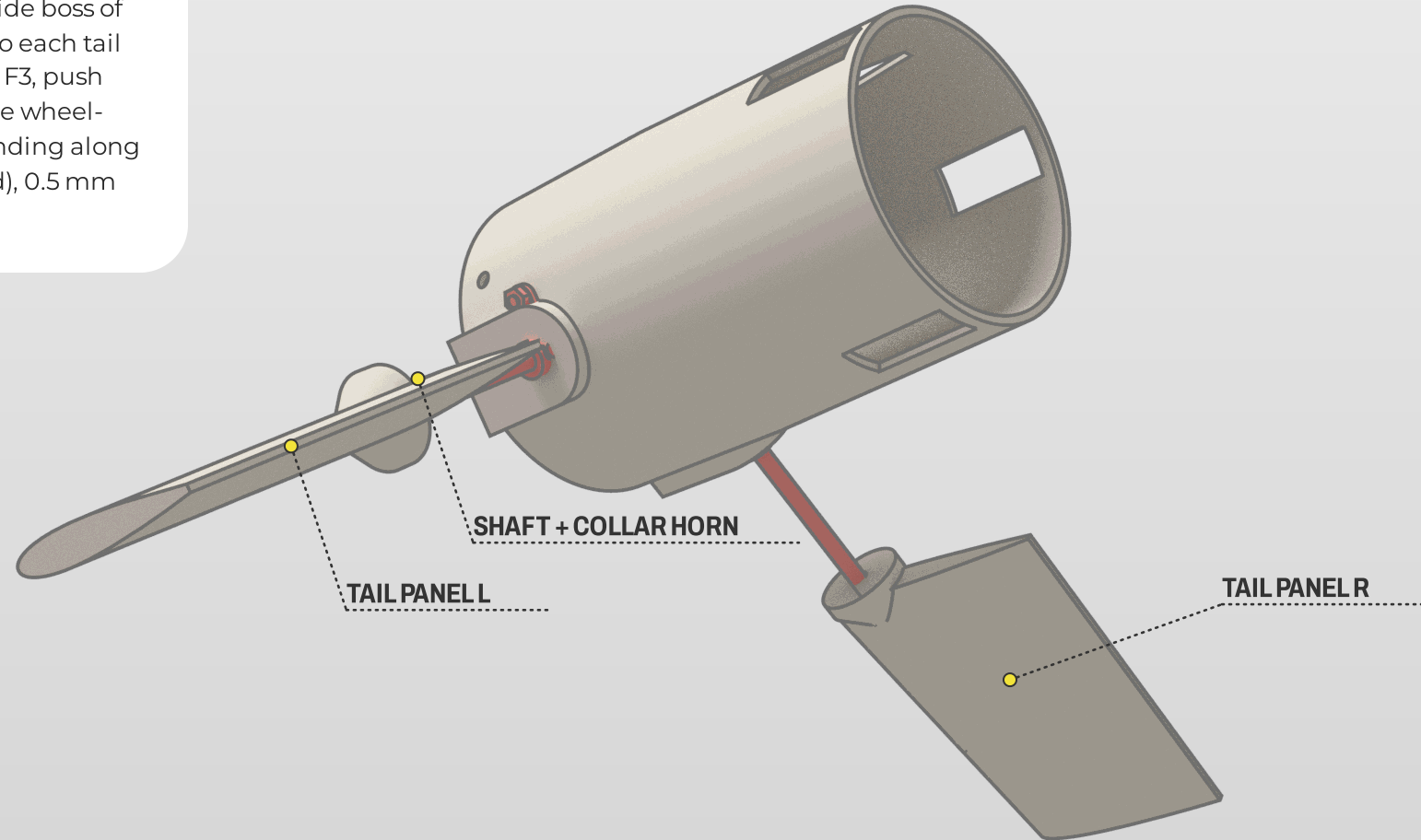


EPOXY

TAIL MODULE



Press a 10 mm brass bushing into each side boss of F3. Epoxy a 2.5 mm steel shaft 35 mm into each tail panel's root bore. From the open front of F3, push each shaft through its bushing and fit the wheel-collar horn with its 8 mm brass plate standing along the panel's normal (not pointing forward), 0.5 mm end play, set screw on a filed flat.

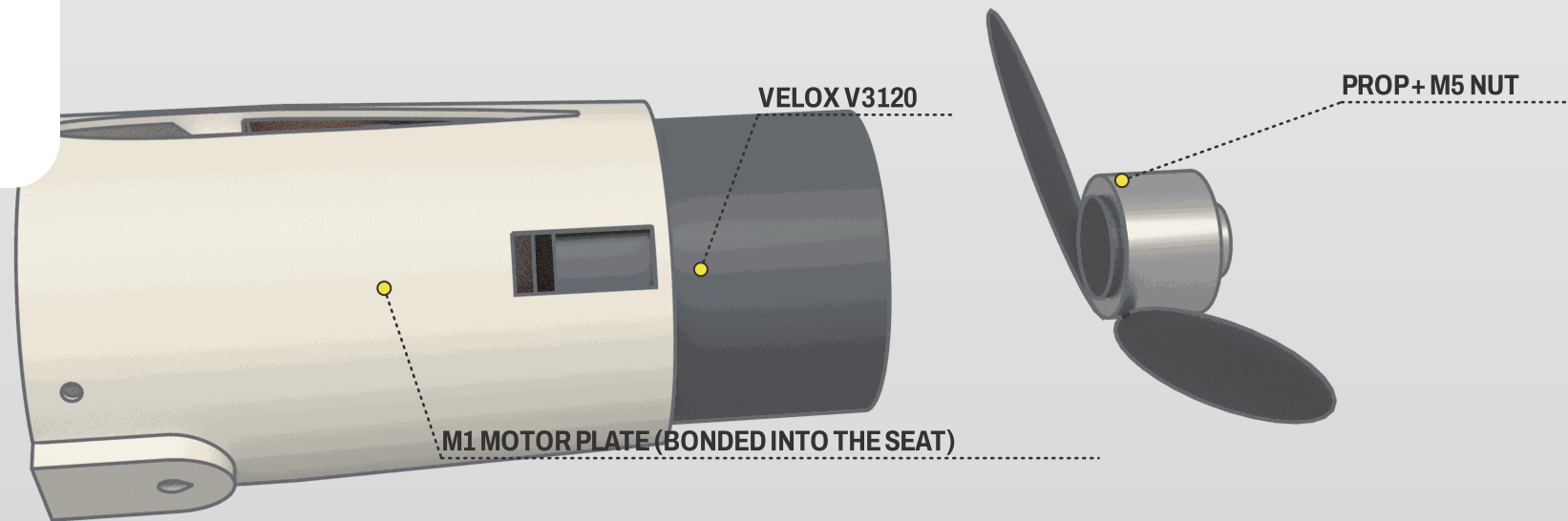


EPOXY

TAIL MODULE



Bond the M1 plate into the conical seat at the rear of F3 with high-temperature epoxy, its edge notches lined up with the cooling windows; it is the motor bulkhead. Then screw the motor to it with its 4 × M3 from inside the module, through the 33 mm opening. Do this before the tail-panel shafts and servos go in. Fit the prop (Custom 2-blade 6 in CNC 7075-T6: P15 (6 × 15, stage 1, ~160-170 m/s) and P18 (6 × 18, stage 2, ~180 m/s)) with the thin M5 nut and thread locker.



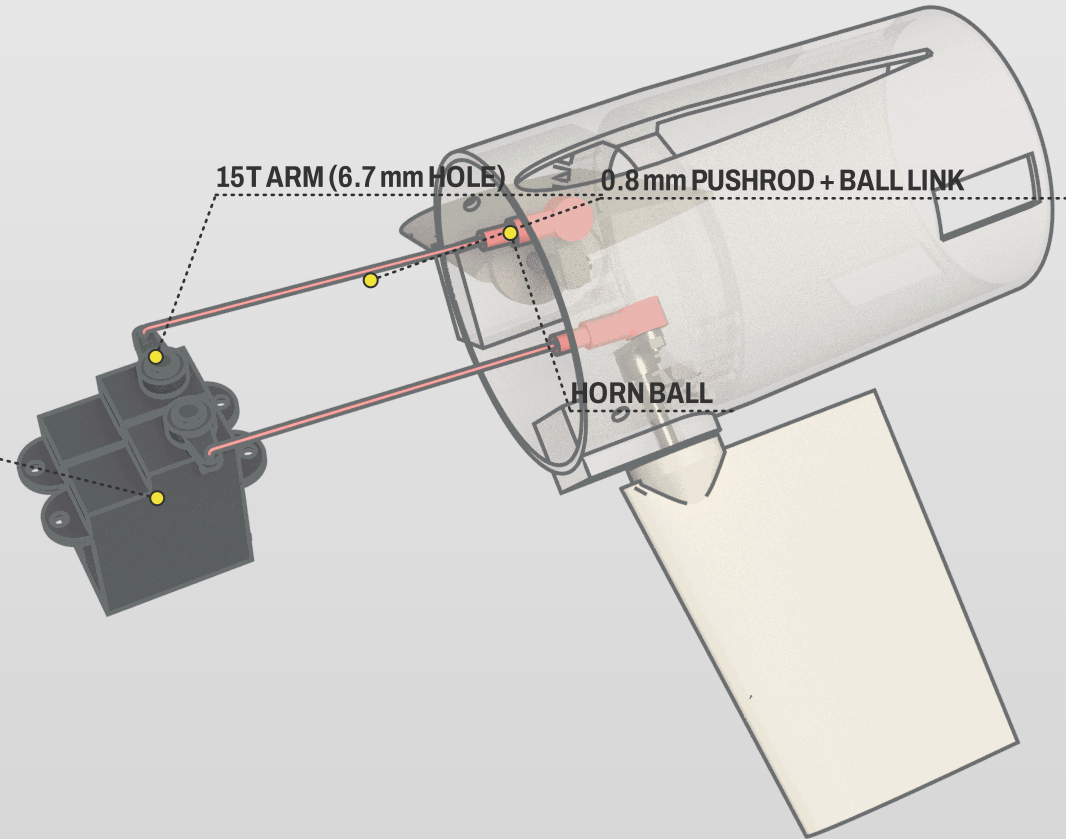
EPOXY

TAIL MODULE



Stand both ES9052MD servos through the cut-outs in the F2 floor (behind the aileron servo) on foam tape, a strap over them, output at the aft end, leads along the side walls. One-sided 15T arm pointing outboard; 0.8 mm music-wire pushrod with a Z-bend in the arm's 6.7 mm hole and a solder sleeve into the micro ball link on the horn's 3 mm ball. Check both panels move freely through $\pm 20^\circ$.

ES9052MD SERVOS

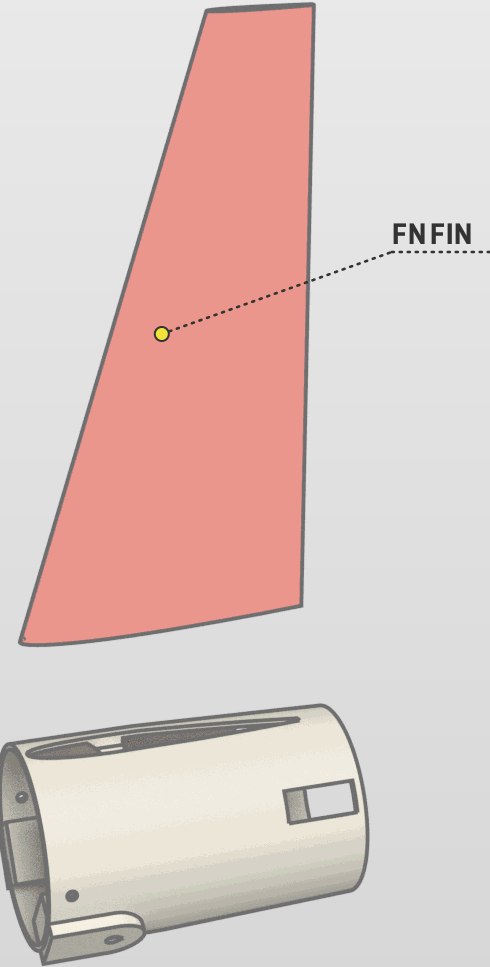


SERVO TAPE

TAIL MODULE



Glue the fin into the top slot of F3 with CA, vertical and square to the wing.

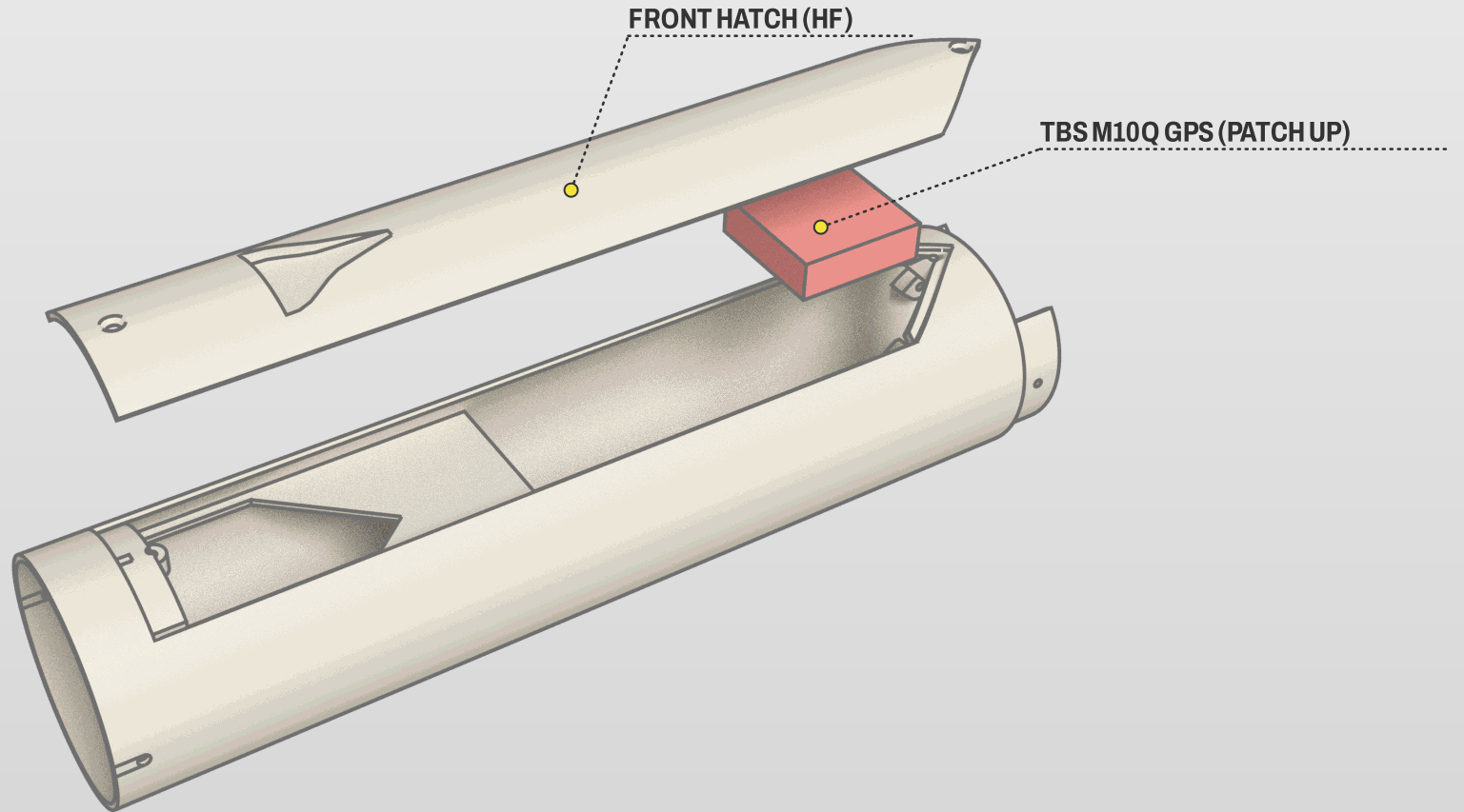


CA GLUE

FUSELAGE ASSEMBLY



Bond the printed GPS carrier (HG) under the rear of the front hatch skin. Foam-tape the TBS M10Q GPS, patch up, to its ceiling, against its rear ledge. Close the hatch with the two M2 × 6 countersunk screws at its ends (they self-tap into the bosses under the lips; do not overtighten). Leave a slack loop in the GPS lead so the hatch lifts off.

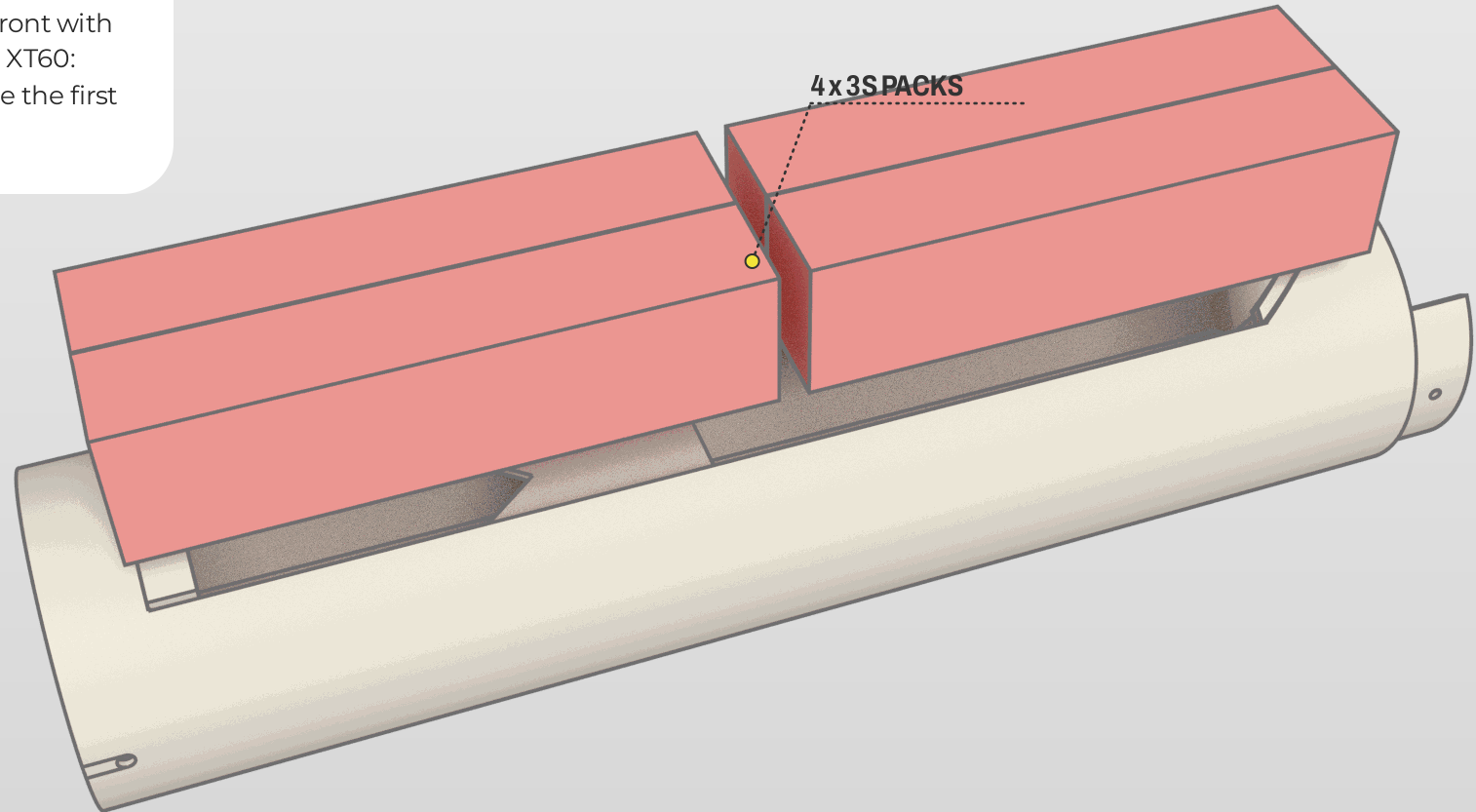


M2 × 6 SCREWS

FUSELAGE ASSEMBLY



Four 3S packs: two side-by-side pairs, one behind the other, on the F1 floor, loaded from the front with the nose off. Series harness in 14 AWG with XT60: measure about 50 V and the polarity before the first connection. Charge each pack on its own.

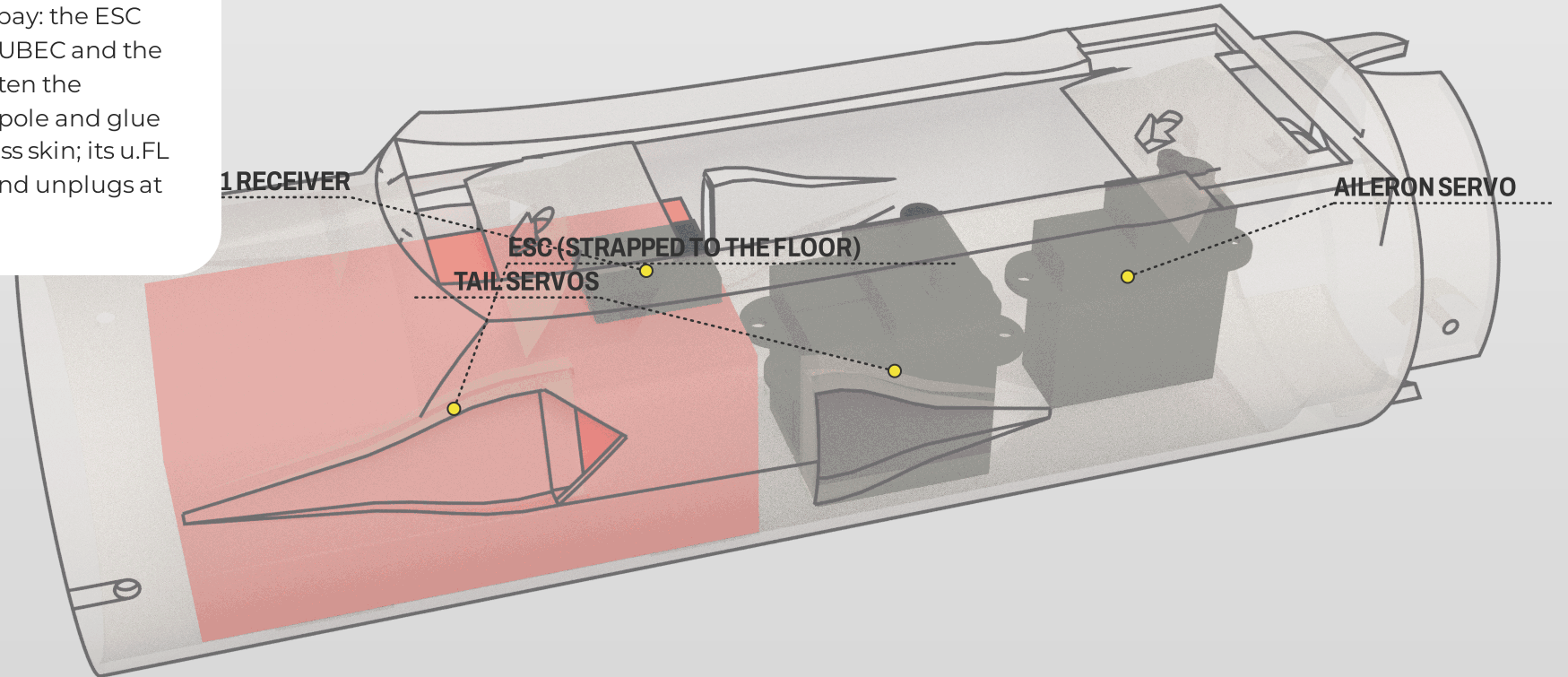


VELCRO STRAP

EQUIPMENT



Matek H743-MINI V3 on a 0.5 mm foam pad on the shelf in F1 above the packs, arrow forward, microSD with the governor script. In the wing bay: the ESC on the lowered floor with a strap, the UBEC and the receiver on it with foam tape. Straighten the receiver's T-antenna into a vertical dipole and glue it in the fin groove before the fin's glass skin; its u.FL extension runs along the F2 ceiling and unplugs at the tail joint.

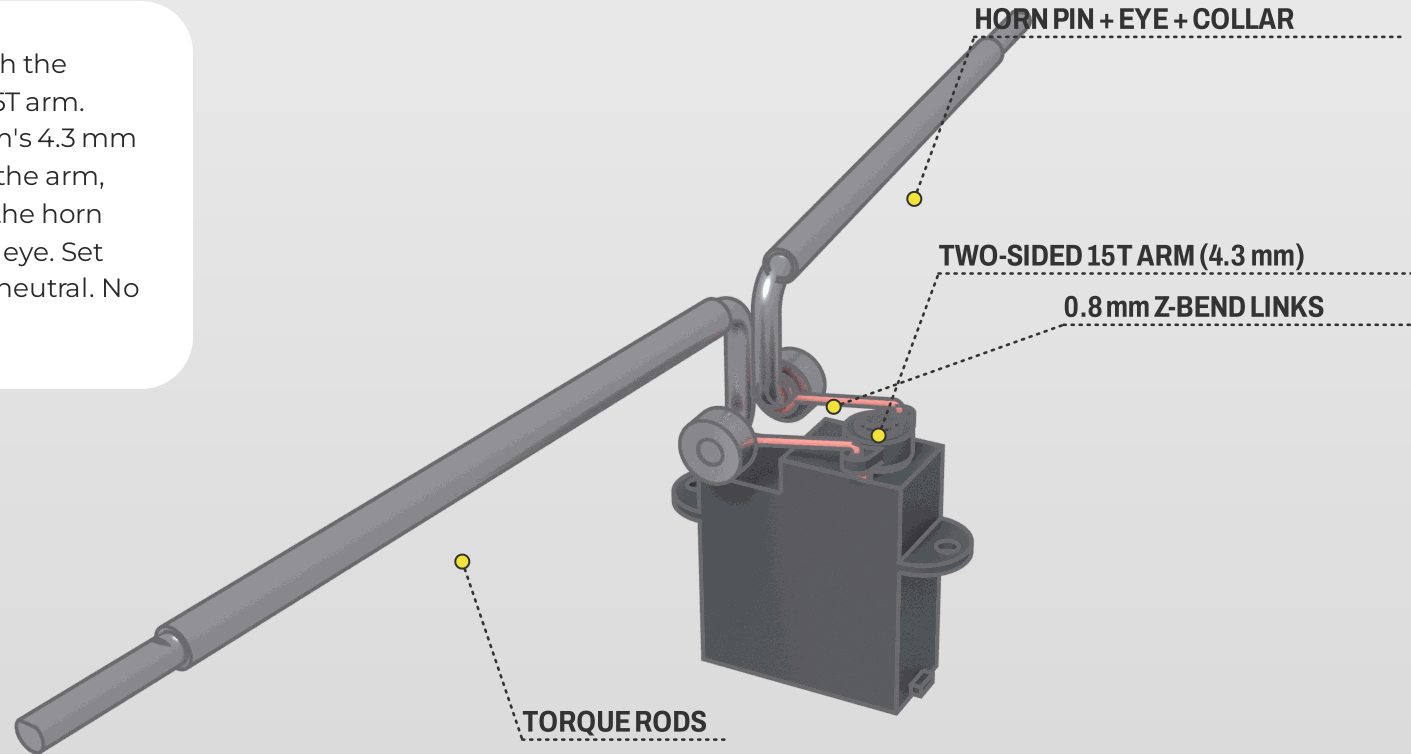


FOAM TAPE

EQUIPMENT



ES9052MD aileron servo standing through the wing-bay floor on foam tape; two-sided 15T arm. Each torque-rod horn connects to the arm's 4.3 mm hole with its own 0.8 mm link: Z-bend in the arm, an eye wrapped round a 2.4 mm drill on the horn pin, a 3/32 in wheel collar outboard of the eye. Set both ailerons 3° trailing edge up at servo neutral. No play at either end.

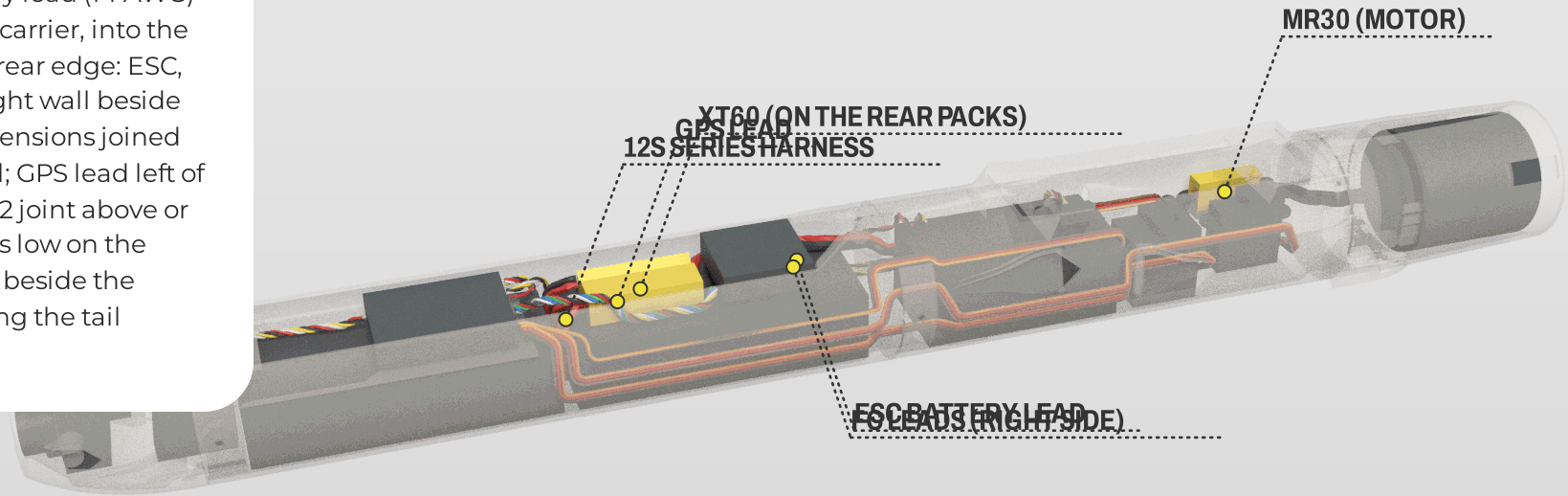


Z-BENDS

EQUIPMENT



12S brick: the four pack leads rise through the 4 mm gap between the pack pairs to the series links and the XT60 lying on the rear pair (plug and unplug it through the front hatch). ESC battery lead (14 AWG) along the right top, beside the GPS carrier, into the ESC's pads. Signal leads to the FC's rear edge: ESC, receiver and UBEC low along the right wall beside the packs; the three servo leads (extensions joined and heat-shrunk) along the left wall; GPS lead left of the XT60. Every lead passes the F1/F2 joint above or below the joint screws. Motor phases low on the right past the tail servo to the MR30 beside the aileron servo: unplug it before pulling the tail module.

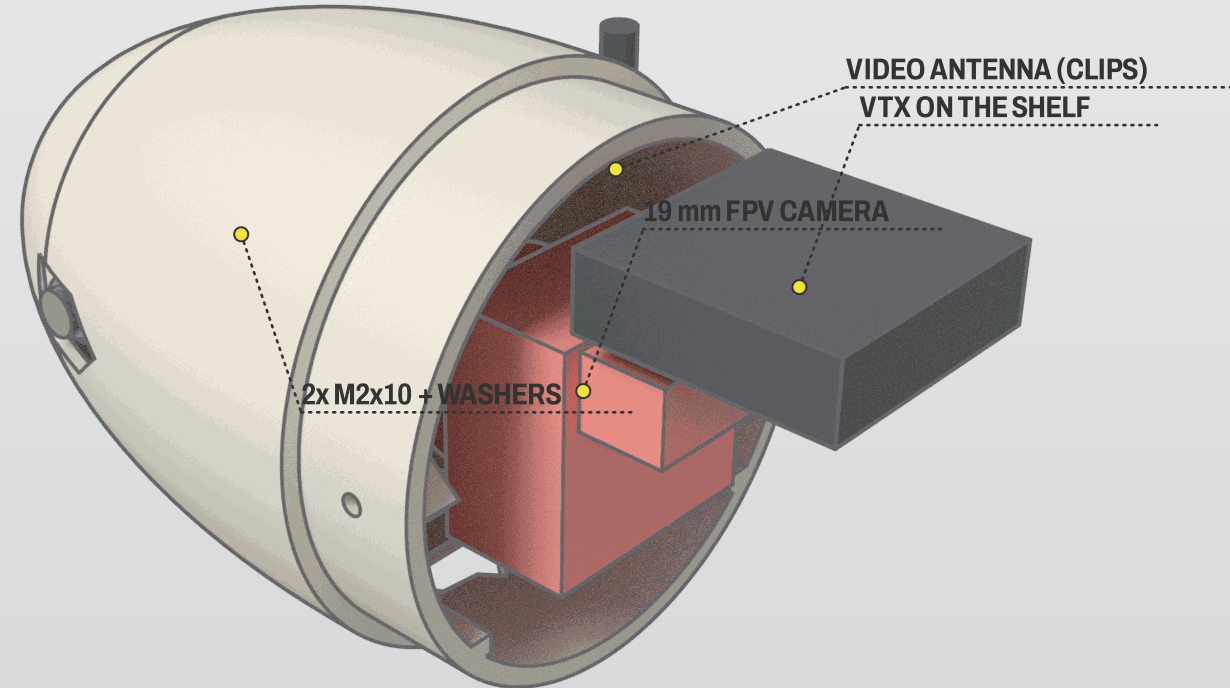


HEAT SHRINK

CAMERA NOSE



Slide the 19 mm camera in from the open rear until the lens reaches the face; 2 × M2 × 10 pan head + 2 × M2 washer (5 mm OD) through the side seats, lens flush, tilt 0–10° up. Fix the VTX to the printed VTX shelf (foam tape + zip tie through its slots), slide the shelf in from the rear onto its two rails and glue it; video antenna in the two clips on the right wall. Plug the one lead at the nose joint. Dome noses: glue the printed dome ring 3 mm into the front of the tube, then fit the clear dome over its lip with canopy glue (no CA).

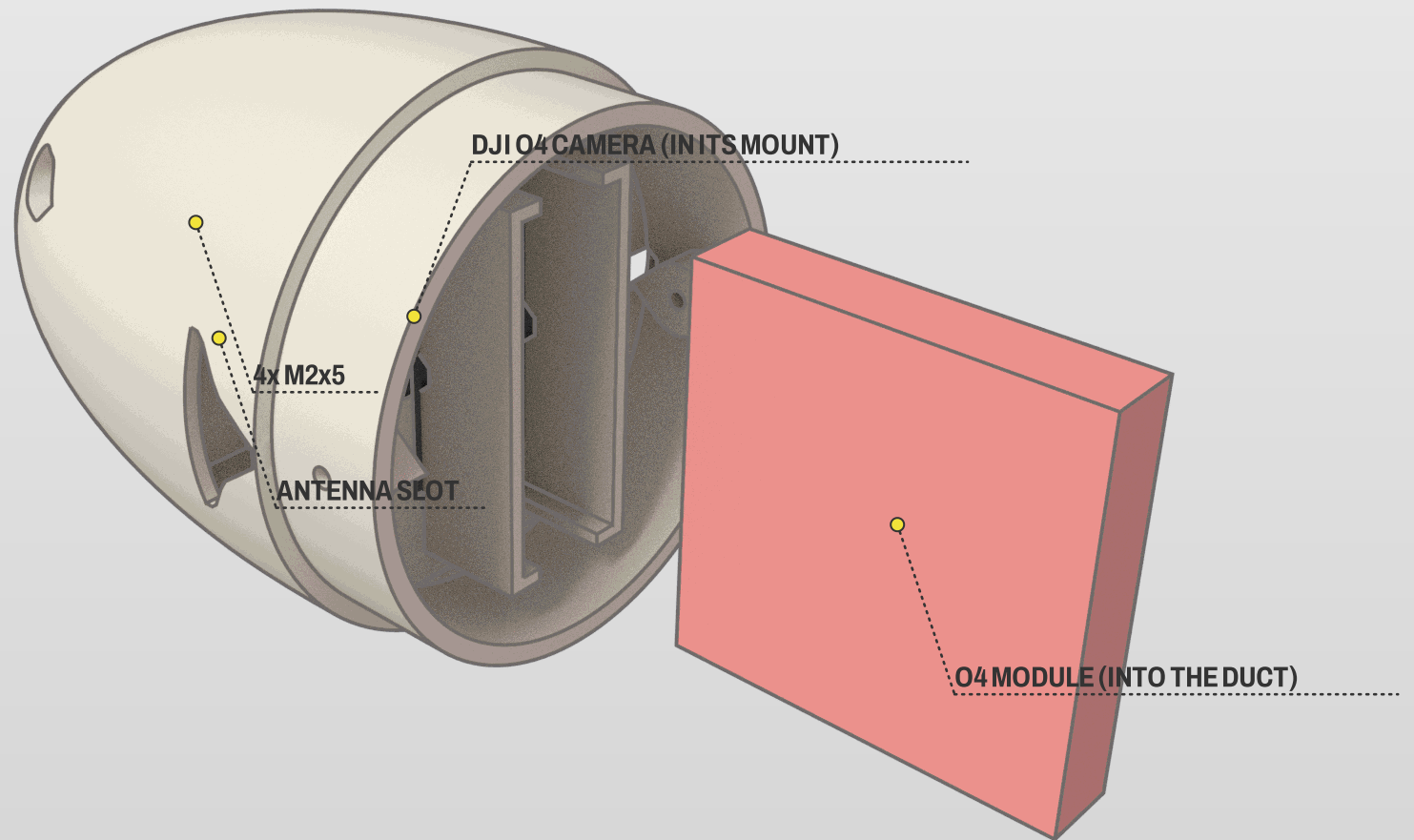


FOAM TAPE

CAMERA NOSE



DJI O4 option: clamp the camera in its mount with 4 × M2 × 5 through the side slots (at most 3.1 mm into the mount). Plug the coax and 3-in-1 lead into the module and slide it edge-first into the duct until it stops; foam strip in the top groove, tape over the rear edge, never over its faces.

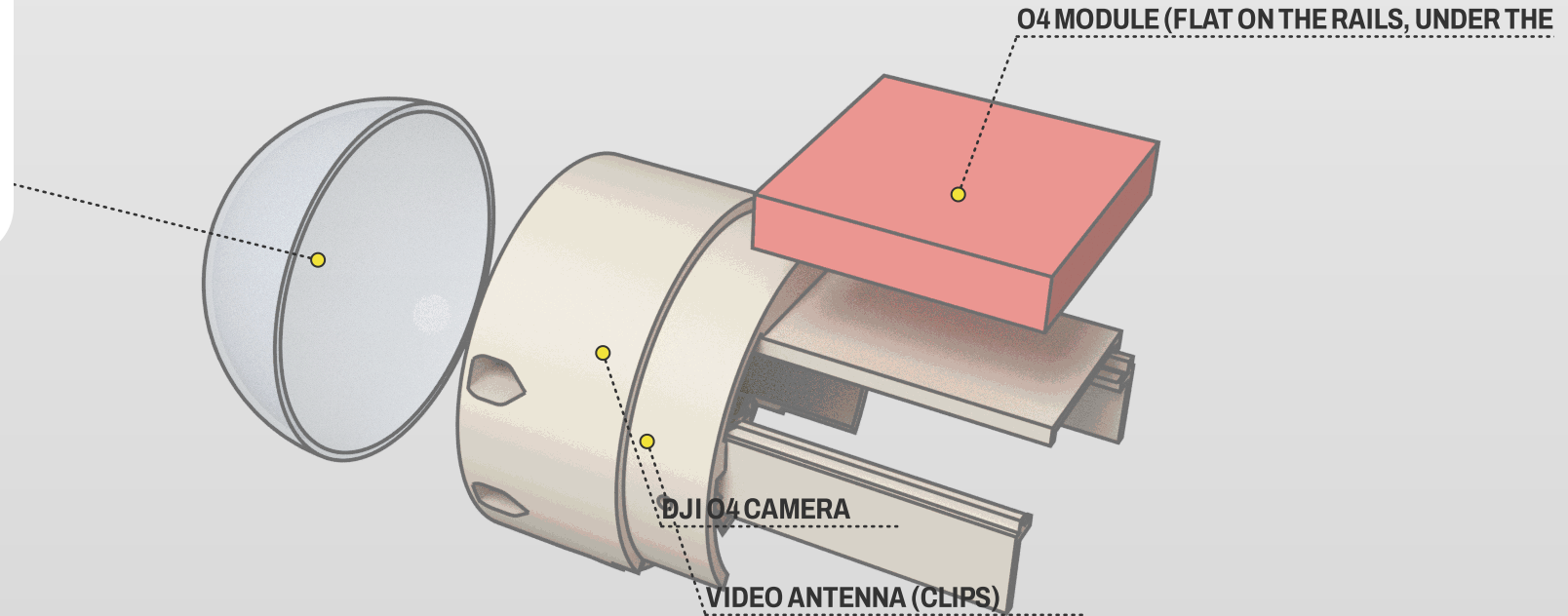


M2 × 5 SCREWS

CAMERA NOSE



DJI O4 dome nose (N00D): clamp the camera as above and clip the antenna on the left wall. Lay the module flat on the two rails behind the spigot, against the front stops, on two strips of thin (0.2 mm) double-sided tape, and slide it under the lid; plug the coax. The 3-in-1 lead goes to the FC right behind it: unplug it at the module before you take the nose off for a pack change. Dome ring and clear dome as on the other dome noses.

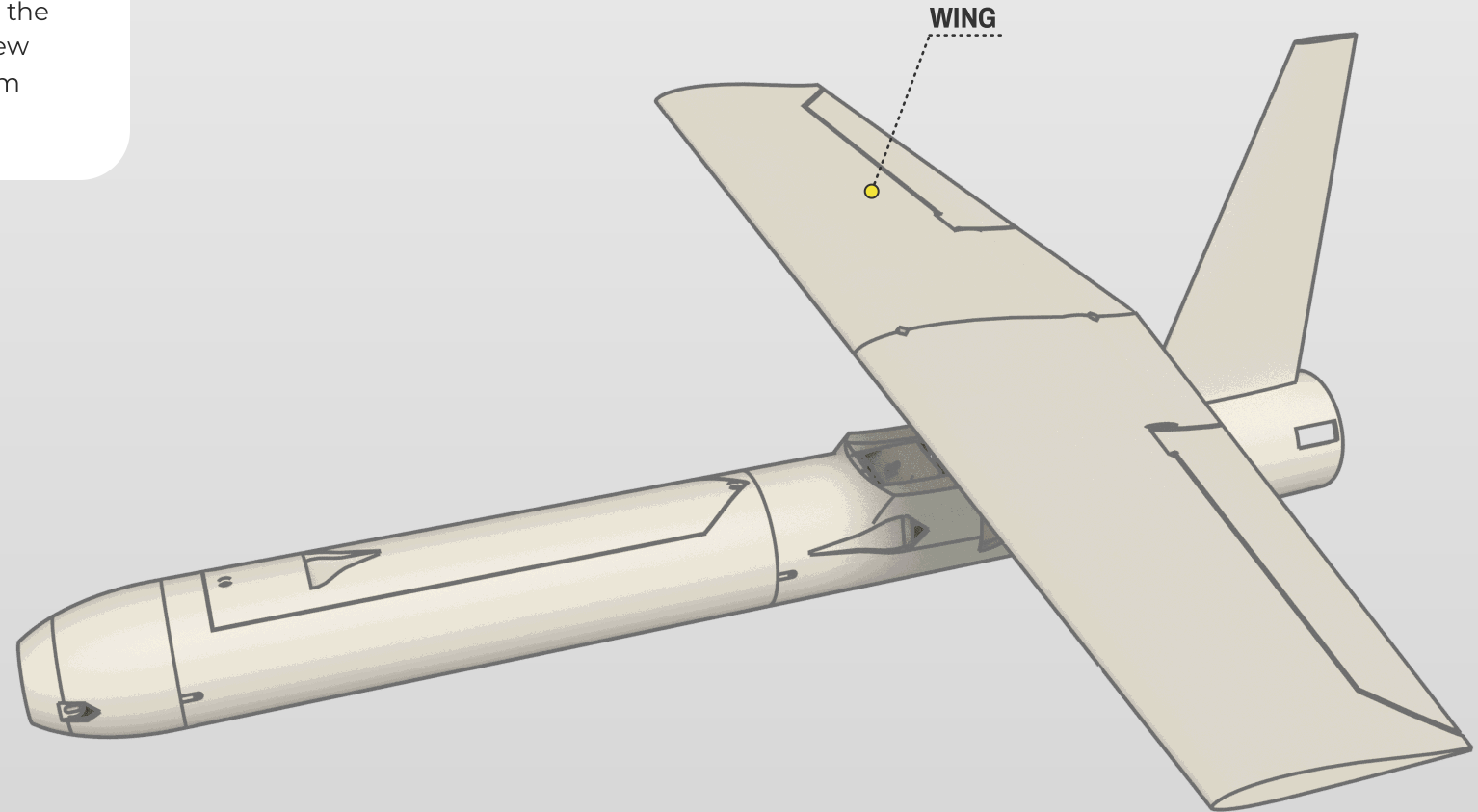


THIN TAPE

FINAL ASSEMBLY



Push an M3 nut up into each bolt-bridge pocket in F2. Set the wing on the saddle and bolt it with the nylon screws: M3 × 18 at the front (a 16 mm screw ends inside the nut), M3 × 8 at the rear (a 12 mm screw reaches the aileron servo arm).



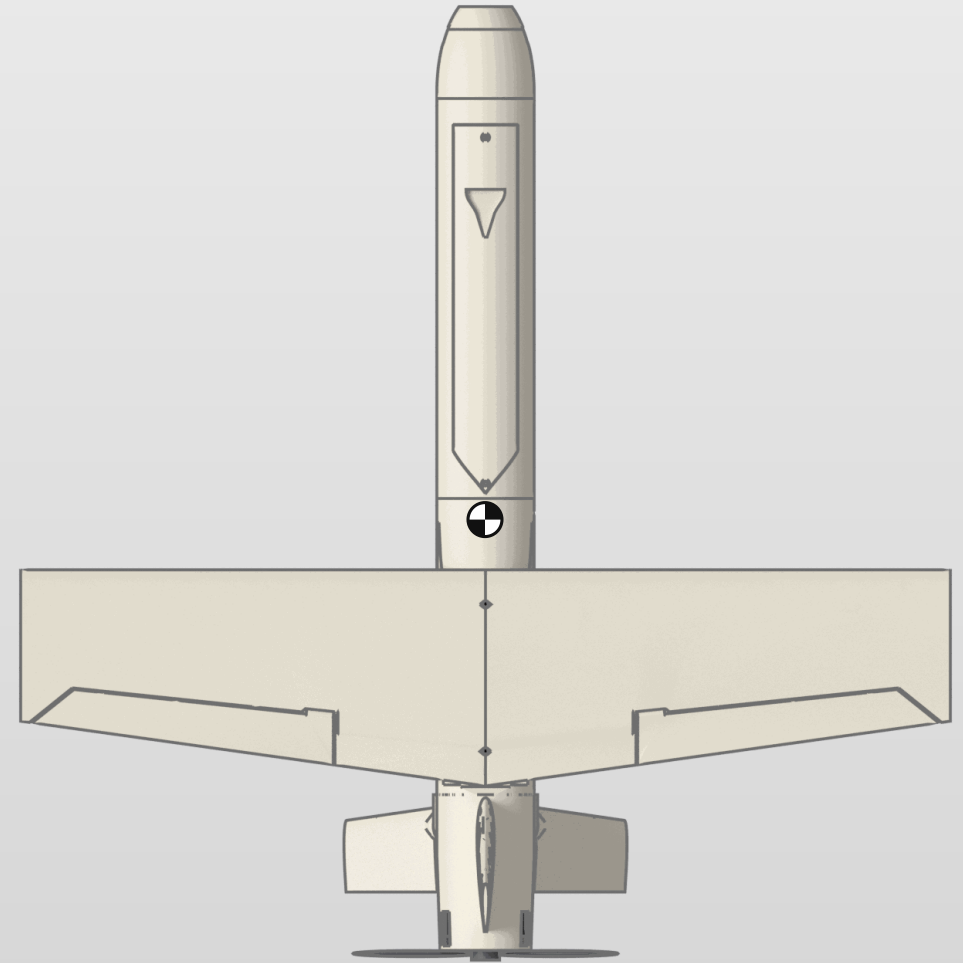
NYLON SCREWS

FINISHING BUILD



Before the first flight: every control surface moves the right way (pitch up: both panel trailing edges up; right roll: right aileron up), throws $\pm 10^\circ$ on the tail panels, about $\pm 10^\circ$ (± 3.5 mm at the trailing edge) on the ailerons, 30 % expo; the prop turns the right way and pushes aft; failsafe set to motor off.

Balance: CG -23 mm behind the wing leading edge at the root (mark it on the wing underside). Slide the battery to trim; for the first flight set the CG 2 mm forward. If a heavy camera needs it, use the nose or tail ballast cup.

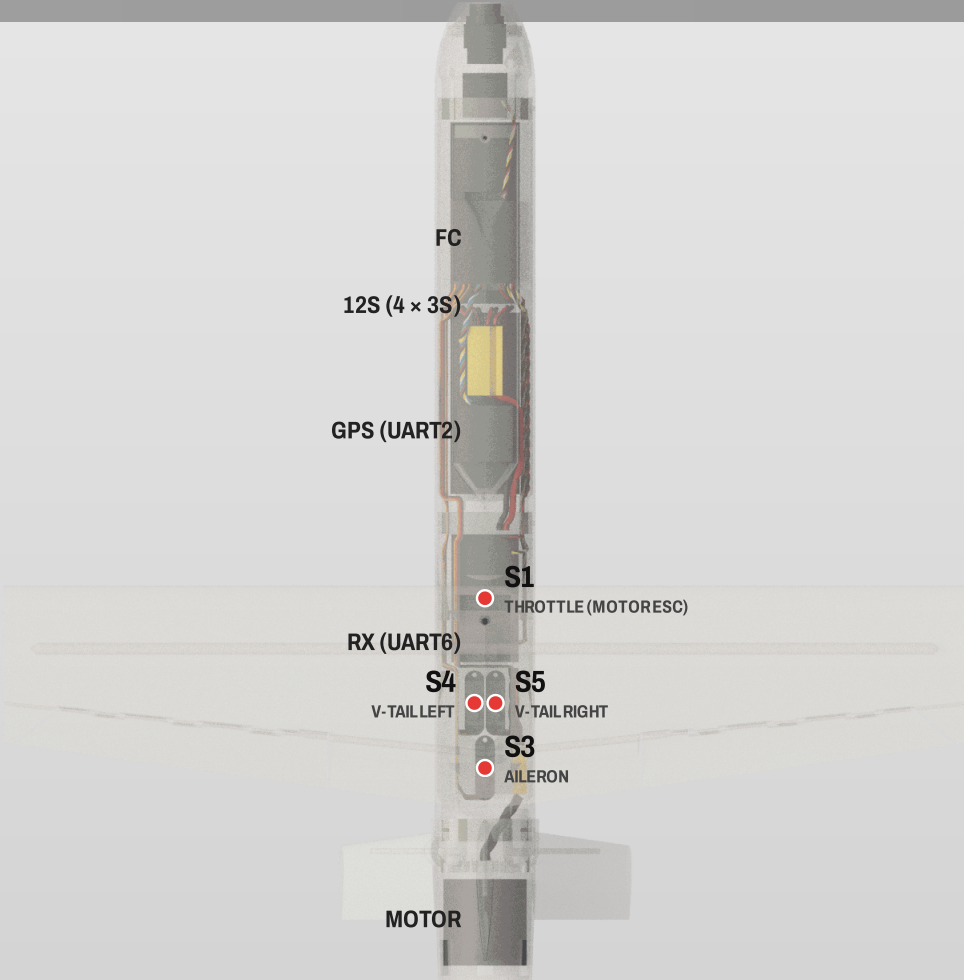


WIRING DIAGRAM



The diagram shows the flight-controller outputs and where each device sits. Signal, power and video leads follow the routes on the wiring page of the assembly guide; the package has this diagram as a separate large page.

PAD	FUNCTION	DEVICE
S1	Throttle (motor ESC)	Sequire SQESC 14130 ESC (bidirectional DShot300)
S2	Unused	Leave empty (DShot group with S1)
S3	Aileron	Aileron servo (ES9052MD, wing bay)
S4	V-tail left	Left tail-panel servo (ES9052MD, in F2 ahead of the tail joint)
S5	V-tail right	Right tail-panel servo (ES9052MD, in F2 ahead of the tail joint)



ARDUPILOT CONFIGURATION

Starting ArduPlane configuration for SkyG Speed 180 on the Matek H743-MINI V3 (firmware MatekH743-bdshot, ArduPlane 4.5 or later), with the rpm-governor script skyg_governor.lua. Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), copy the script to the microSD card, reboot, load the .param file again (for the GOV_ parameters), then work through the bench checks.

Flight controller: on the shelf above the battery in F1, arrow forward, on a 0.5 mm foam pad; 5 V from the UBEC; servo rail 6 V from the UBEC.

Firmware: ArduPlane 4.5 or later · **Board:** Matek H743-MINI V3 (ArduPilot firmware: MatekH743-bdshot)

[The setup document and the starting .param file are in the package \(ArduPilot folder\). Verify every value on the bench before flight.](#)

SKY G

ARDUPILOT CONFIG

SKYG SPEED 180

MATEK H743-MINI V3

FLIGHT TEST AND SPEED STAGES



The speed envelope opens in stages. Fly each stage only after the ground checks and the previous stage pass: thrust-stand run, wing load and twist test, control freeplay ≤ 0.2 mm (≤ 0.05 mm above 45 m/s), aileron hinge stiffness, and the tap test. At each run give short aileron and pitch doublets; any buzz or vibration: throttle to zero, gentle pull-up, land and inspect.

Full procedure, limits and log sheet: the test plan in the Engineering folder.

Stage Vne	GOV_RPM (full stick)	Expected level speed
30 m/s	5,900 rpm	29 m/s
35 m/s	6,700 rpm	34 m/s
40 m/s	7,500 rpm	40 m/s
45 m/s	8,300 rpm	44 m/s
50 m/s	9,100 rpm	49 m/s
55 m/s	10,000 rpm	54 m/s
60 m/s	10,800 rpm	59 m/s
65 m/s	11,600 rpm	64 m/s
70 m/s	12,400 rpm	69 m/s
75 m/s	13,200 rpm	74 m/s
80 m/s	14,000 rpm	79 m/s
85 m/s	14,800 rpm	84 m/s
90 m/s	15,700 rpm	89 m/s
95 m/s	16,200 rpm	92 m/s
100 m/s	16,200 rpm	92 m/s
105 m/s	16,200 rpm	92 m/s
110 m/s	18,900 rpm	109 m/s
115 m/s	19,700 rpm	114 m/s
120 m/s	20,500 rpm	119 m/s
125 m/s	21,300 rpm	124 m/s
130 m/s	22,100 rpm	129 m/s
135 m/s	22,900 rpm	134 m/s
140 m/s	23,700 rpm	139 m/s
145 m/s	24,500 rpm	144 m/s
150 m/s	25,300 rpm	149 m/s
155 m/s	22,600 rpm	153 m/s

SKYG

SKYG SPEED 180

